

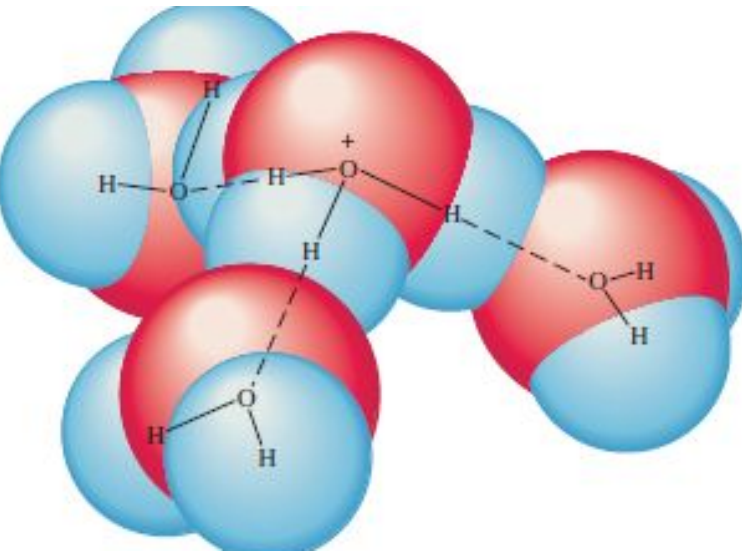


Red cabbage juice is green in basic solution (first beaker), but turns red when dry ice, or solid CO_2 , is added (center beaker) to give an acidic solution (last beaker).

Acid-Base Concepts

- **Antoine Lavoisier** was one of the first chemists to try to explain what makes a substance acidic. In 1777 he proposed that **oxygen** was an essential element in acids. (Oxygen, which he named, means “**acid-former**” in **Greek**.)
- But in 1808 **Humphry Davy** showed that hydrogen chloride, which dissolves in water to give hydrochloric acid, contains only hydrogen and chlorine.
- Although some chemists argued that chlorine was a compound of oxygen, chlorine was eventually proved to be an element.
- Chemists then noted that hydrogen, not oxygen, must be the essential constituent of acids. The cause of acidity and basicity was first explained in 1884 by **Svante Arrhenius**.

Arrhenius Concept of Acids and Bases



The hydronium ion, H_3O^+
The species is shown here
hydrogen bonded to three water
molecules. The positive charge
shown is actually distributed
over the ion.

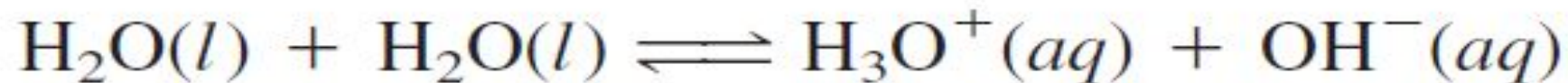
An acid is a substance that, when dissolved in water, increases the concentration of hydronium ion, $\text{H}_3\text{O}^+(\text{aq})$.

For simplicity, chemists frequently use the notation $\text{H}^+(\text{aq})$ for the $\text{H}_3\text{O}^+(\text{aq})$ ion, and call it the hydrogen ion. of water molecules. One such species is shown in Figure

A base is a substance that, when dissolved in water, increases the concentration of hydroxide ion, $\text{OH}^-(\text{aq})$.

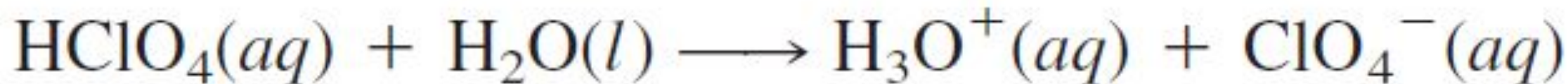
Arrhenius Concept of Acids and Bases

The special role of the **hydronium ion (or hydrogen ion) and the hydroxide ion** in aqueous solutions arises from the following reaction:



The addition of acids and bases alters the concentrations of these ions in water.

In **Arrhenius's theory**, a strong acid is a substance that **completely ionizes** in **aqueous solution to** give **$\text{H}_3\text{O}^+(aq)$** and an anion. An example is perchloric acid, HClO_4 .



Other examples of strong acids are H_2SO_4 , HI , HBr , HCl , and HNO_3 .

Strong base **completely ionizes** in **aqueous solution** to give **OH⁻** and a cation.

Sodium hydroxide is an example of a strong base.



Common Strong Acids and Bases

Strong Acids



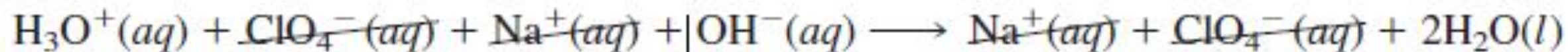
Strong Bases*



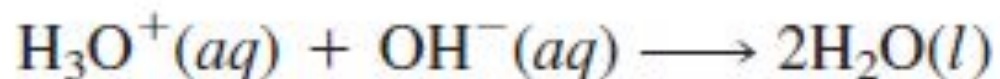
*In general, the Group IA and Group IIA hydroxides (except beryllium hydroxide) are strong bases.

Evidence for the Arrhenius theory comes from the heat of reaction, ΔH° , for the neutralization of **a strong acid by a strong base**. This neutralization is essentially the reaction of $\text{H}_3\text{O}^+(\text{aq})$ with $\text{OH}^-(\text{aq})$ and should therefore always give the same ΔH° per mole of water formed.

For example, if you write the neutralization of HClO_4 with NaOH in ionic form, you have



After canceling, you get the net ionic equation



Experimentally, it is found that all neutralizations involving strong acids and bases have the same **ΔH° : -55.90 kJ per mole of H^+** . This indicates that the same reaction occurs in each neutralization, as Arrhenius's theory predicts.

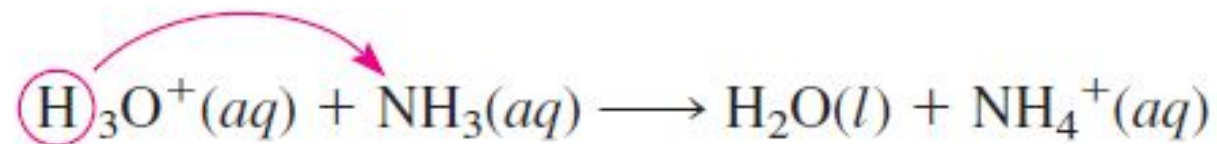
Brønsted-Lowry Concept of Acids and Bases

In 1923 the Danish chemist **Johannes N. Brønsted** (1879–1947) and, independently, the British chemist **Thomas M. Lowry** (1874–1936) pointed out that acid–base reactions can be seen as proton-transfer reactions and that acids and bases can be defined in terms of this proton (H^+) transfer. According to the **Brønsted–Lowry concept**, an *acid is the species donating a proton in a proton-transfer reaction*. *A base is the species accepting the proton in a proton-transfer reaction*

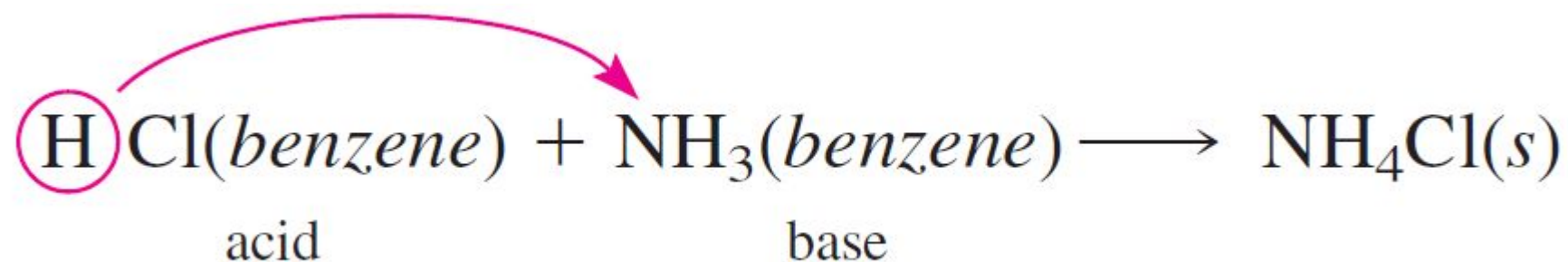
Consider, for example, the reaction of hydrochloric acid with ammonia.



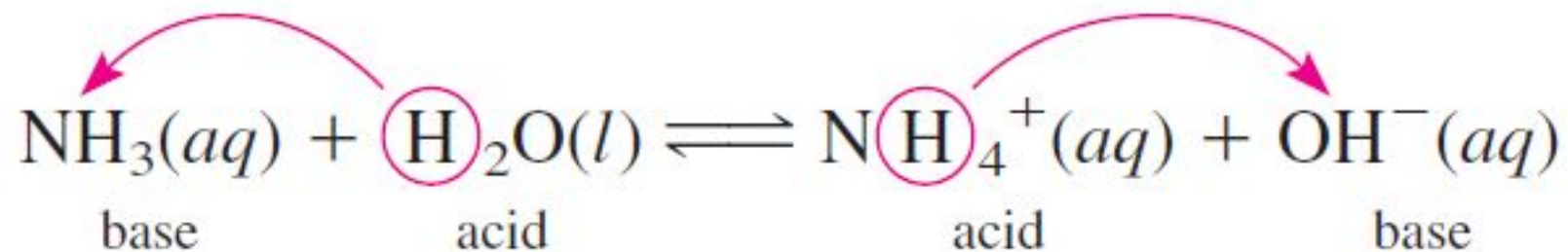
After canceling Cl^- , you obtain the net ionic equation.

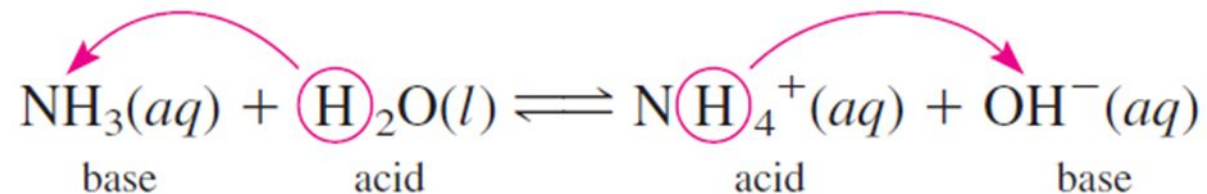


The Brønsted–Lowry concept to the reaction of HCl and NH₃ dissolved in non-aqueous media benzene, C₆H₆. In benzene, HCl and NH₃ are not ionized. The equation is



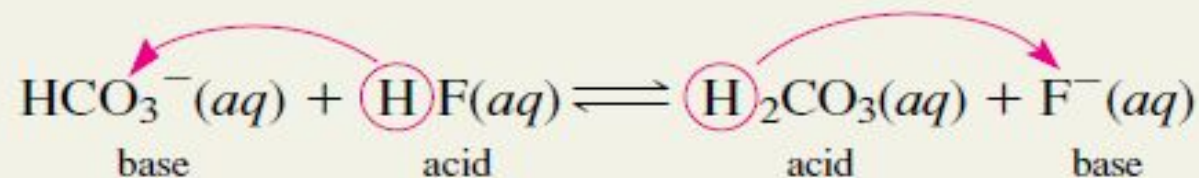
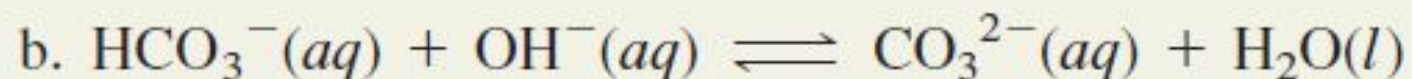
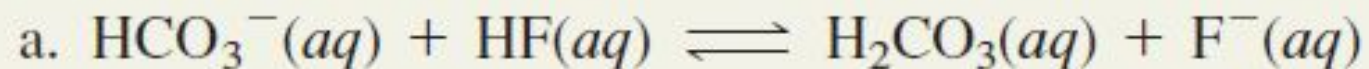
In any reversible acid–base reaction, both forward and reverse reactions involve proton transfers. Consider the reaction of NH₃ with H₂O.





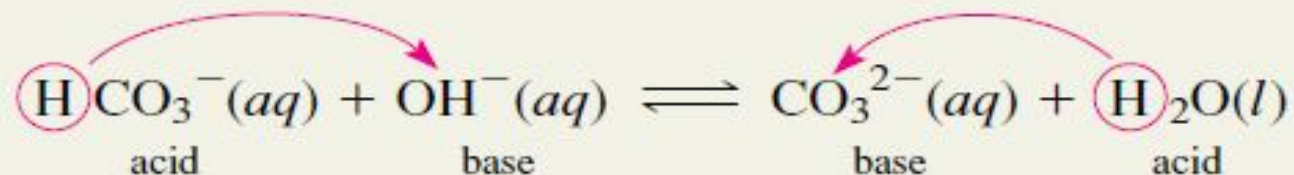
- Note that NH_3 and NH_4^+ differ by **a proton**.
- That is, **NH_3 becomes the NH_4^+ ion** by gaining a proton, whereas the **NH_4^+ ion becomes the NH_3 molecule** by losing a proton.
- The species **NH_4^+ and NH_3 are a conjugate acid–base pair**. *A conjugate acid–base pair consists of two species in an acid–base reaction, one acid and one base, that differ by the loss or gain of a proton.*
- The acid in such a pair is called the *conjugate acid of the base*, whereas the *base is the conjugate base of the acid*. Here NH_4^+ is the conjugate acid of NH_3 , and NH_3 is the conjugate base of NH_4^+ .

In the following equations, label each species as an acid or a base. Show the conjugate acid–base pairs.



In this reaction, H_2CO_3 and HCO_3^- are a conjugate acid–base pair, as are HF and F^- .

b. You have



Here HCO_3^- and CO_3^{2-} are a conjugate acid–base pair, as are H_2O and OH^- . Note that although HCO_3^- functions as an acid in this reaction, it functions as a base in a.

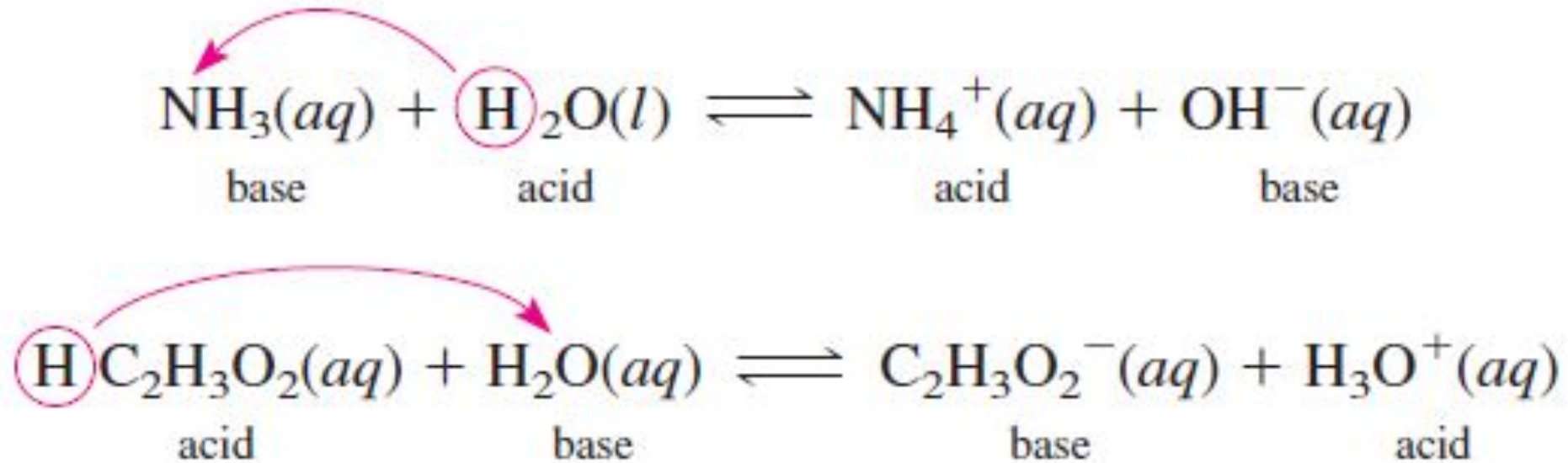


label each species as an acid or a base. For the base on the left, what is the conjugate acid?

An amphiprotic species

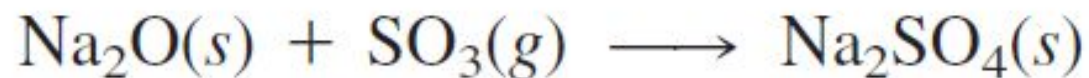
A species that can act as either an acid or a base (it can lose or gain a proton), depending on the other reactant. For example, HCO_3^- acts as an acid in the presence of OH^- but as a base in the presence of HF . Anions with ionizable hydrogens, such as HCO_3^- , and certain solvents, such as water, are amphiprotic.

The amphoteric characteristic of water is important in the acid–base properties of aqueous solutions. Consider, for example, the reactions of water with the base NH_3 and with the acid $\text{HC}_2\text{H}_3\text{O}_2$ (acetic acid).



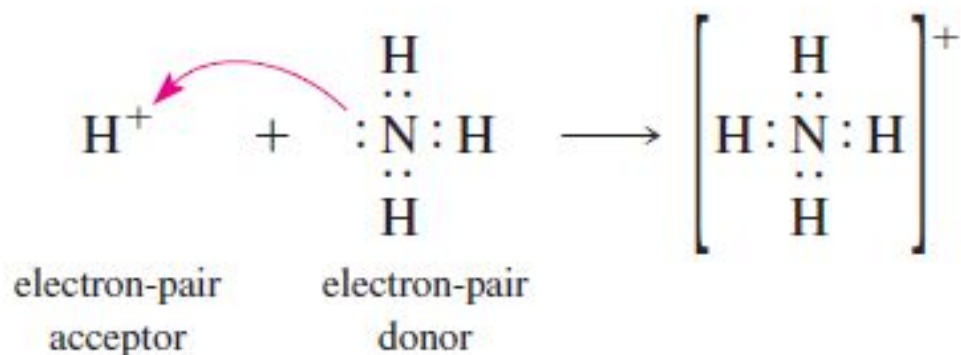
Lewis Concept of Acids and Bases

Certain reactions have the characteristics of acid–base reactions but do not fit the **Brønsted–Lowry concept**. An example is the reaction of the basic oxide **Na₂O** with the acidic oxide **SO₃** to give the salt **Na₂SO₄**



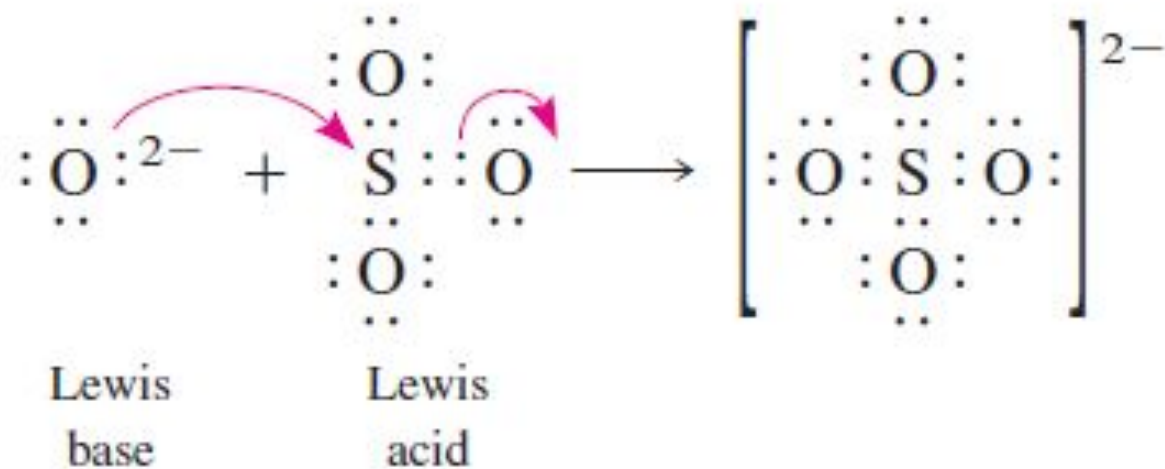
A Lewis acid is a species that can form a covalent bond by accepting an electron pair from another species; a Lewis base is a species that can form a covalent bond by donating an electron pair to another species.

Consider again the neutralization of NH₃ by HCl in aqueous solution, mentioned earlier. It consists of the reaction of a proton from H₃O⁺ with NH₃:



Here the red arrow shows the proton accepting an electron pair from NH₃ and an H—N bond being formed. The proton is an electron-pair acceptor, so it is a Lewis acid. Ammonia, NH₃, which has a lone pair of electrons, is an electron-pair donor and therefore a Lewis base.

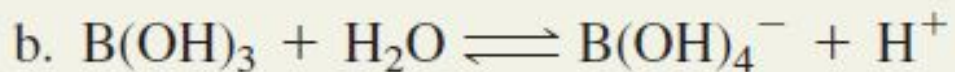
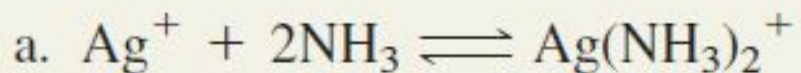
Now let us look at the reaction of Na_2O with SO_3 . It involves the reaction of the oxide ion, O^{2-} from the ionic solid, Na_2O , with SO_3 .



Here SO_3 accepts the electron pair from the O^{2-} ion. At the same time, *an electron pair from the $\text{S}=\text{O}$ bond moves to the O atom*. Thus, O^{2-} is the Lewis base and SO_3 is the Lewis acid.

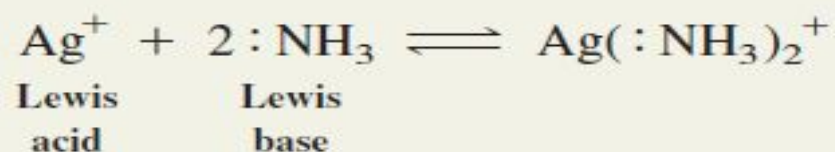
Identifying Lewis Acid and Base Species

In the following reactions, identify the Lewis acid and the Lewis base.

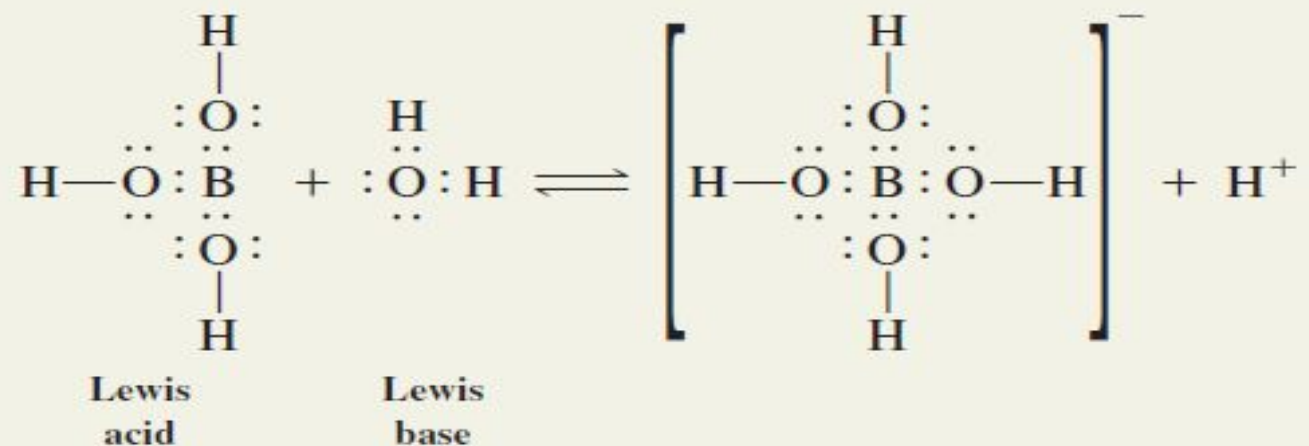


Write the equations using Lewis electron-dot formulas. Then identify the electron-pair acceptor, or Lewis acid, and the electron-pair donor, or Lewis base.

a. The silver ion, Ag^+ , forms a complex ion with two NH_3 molecules.



b. The reaction is



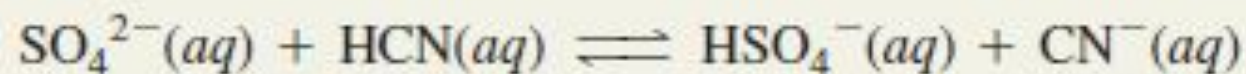
Relative Strengths of Acids and Bases

	Acid	Base	
Strongest acids	HClO ₄	ClO ₄ ⁻	Weakest bases
	H ₂ SO ₄	HSO ₄ ⁻	
	HI	I ⁻	
	HBr	Br ⁻	
	HCl	Cl ⁻	
	HNO ₃	NO ₃ ⁻	
	H ₃ O ⁺	H ₂ O	
	HSO ₄ ⁻	SO ₄ ²⁻	
	H ₂ SO ₃	HSO ₃ ⁻	
	H ₃ PO ₄	H ₂ PO ₄ ⁻	
	HNO ₂	NO ₂ ⁻	
	HF	F ⁻	
	HC ₂ H ₃ O ₂	C ₂ H ₃ O ₂ ⁻	
	Al(H ₂ O) ₆ ³⁺	Al(H ₂ O) ₅ OH ²⁺	
	H ₂ CO ₃	HCO ₃ ⁻	
	H ₂ S	HS ⁻	
	HCIO	CIO ⁻	
	HBrO	BrO ⁻	
	NH ₄ ⁺	NH ₃	
	HCN	CN ⁻	
	HCO ₃ ⁻	CO ₃ ²⁻	
	H ₂ O ₂	HO ₂ ⁻	
	HS ⁻	S ²⁻	
Weakest acids	H ₂ O	OH ⁻	Strongest bases

An acid is strong if it completely ionizes

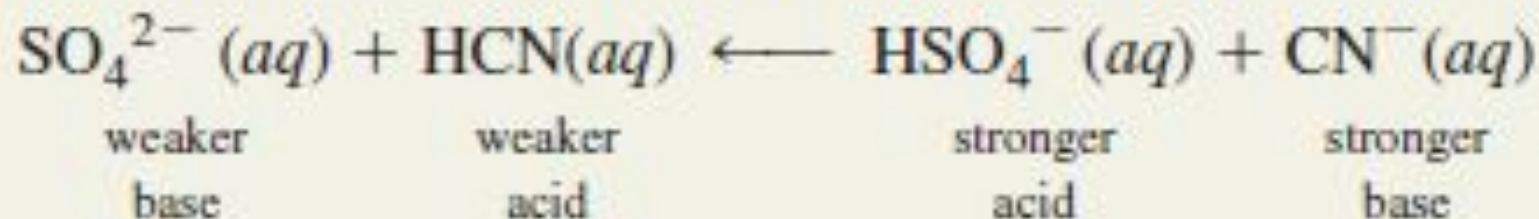
Deciding Whether Reactants or Products Are Favored in an Acid–Base Reaction

For the following reaction, decide which species (reactants or products) are favored at the completion of the reaction.



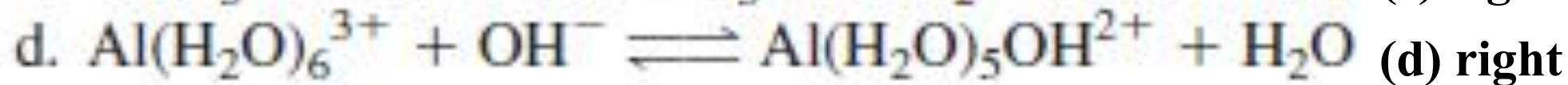
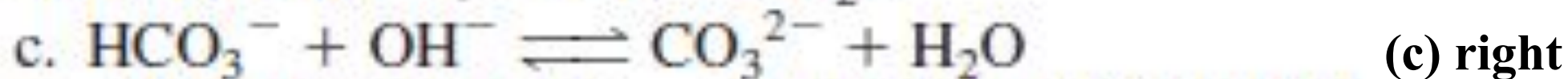
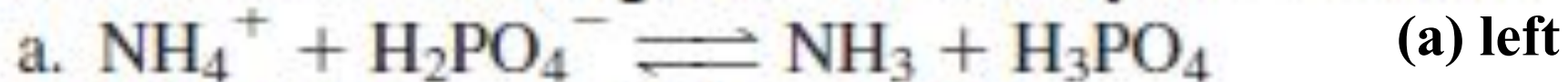
Use Table 16.2 to compare the relative strengths of acids and bases.

If you compare the relative strengths of the two acids HCN and HSO_4^{-} , you see that HCN is weaker. Or, comparing the bases SO_4^{2-} and CN^{-} , you see that SO_4^{2-} is weaker. Hence, the reaction would normally go from right to left.



The reactants are favored.

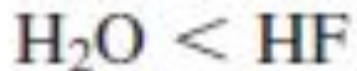
16.41 Use Table 16.2 to decide whether the species on the left or those on the right are favored by the reaction.



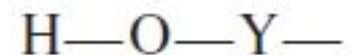
Consider a series of binary acids, HX, formed from a given column of elements of the periodic table. The acids would be compounds of these elements with hydrogen, such as the binary acids of the Group VIIA elements: HF, HCl, HBr, and HI. As you go down the column of elements, each time adding a shell of electrons to the atom, the radius increases markedly. For this reason, the size of atom X is the dominant factor in determining the acid strength. In going down a column of elements of the periodic table, the size of atom X increases, the H—X bond strength decreases, and the strength of the binary acid increases. You predict the following order of acid strength



Now the polarity of the H—X bond becomes the dominant factor in determining acid strength. Going across a row of elements of the periodic table, the electronegativity increases, the H—X bond *polarity increases, and the acid strength increases*. For example, the binary acids of the last two elements of the second period are H₂O and HF. The acid strengths are

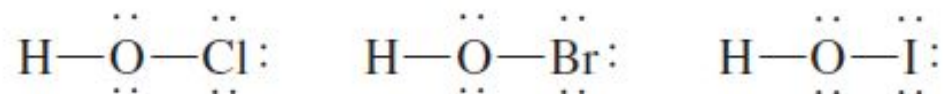


Now consider the oxoacids. An oxoacid has the structure



The acidic H atom is always attached to an O atom, which, in turn, is attached to an atom Y. Other groups, such as O atoms or O—H groups, may also be attached to Y. Bond polarity appears to be the dominant factor determining relative strengths of the oxoacids. This, in turn, depends on the electronegativity of atom Y.

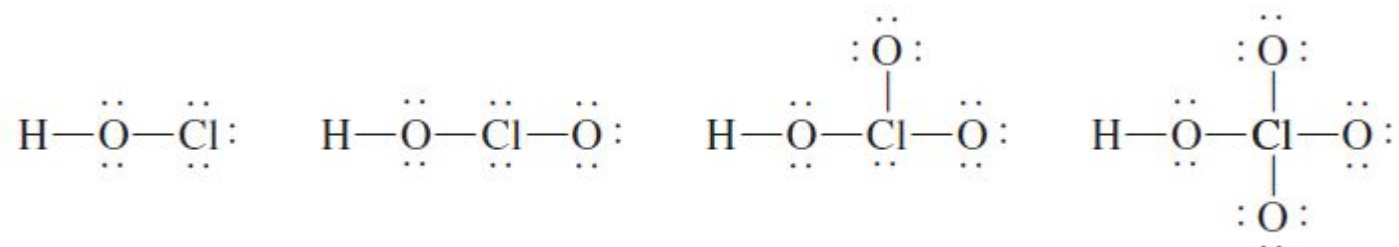
If the electronegativity of atom Y is large, the H—O bond is relatively polar and the acid strength large. For a series of oxoacids of the same structure, differing only in the atom Y, the acid strength increases with the electronegativity of Y. Consider, for example, the acids HClO, HBrO, and HIO.



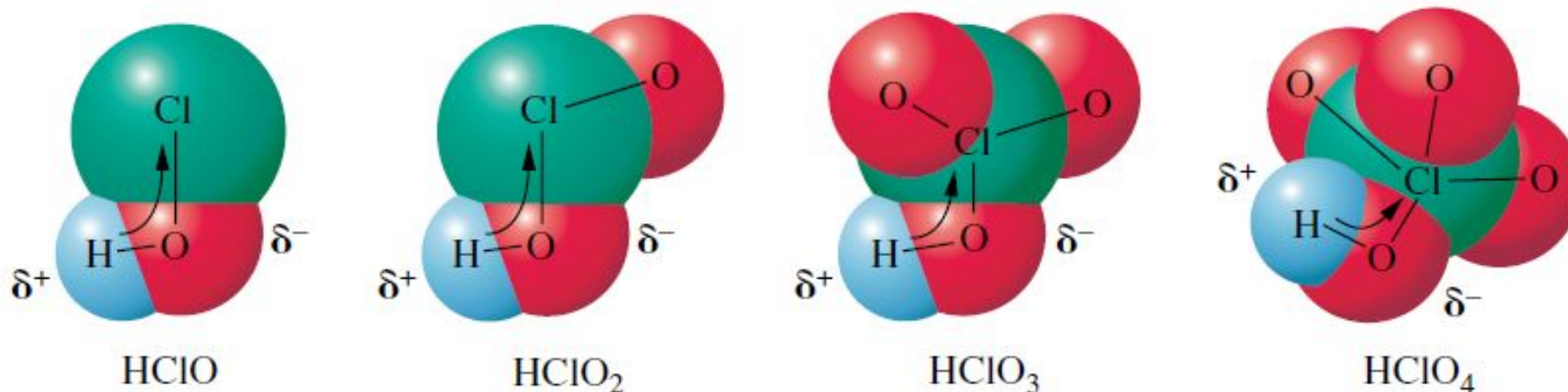
The electronegativity of Group VIIA elements decreases going down the column of elements, so the electronegativity decreases from Cl to Br to I. The order of acid strengths is



For a series of oxoacids, $(\text{HO})_m\text{YO}_n$, the acid strength increases with n , the number of O atoms bonded to Y (excluding O atoms in OH groups). The oxoacids of chlorine provide an example

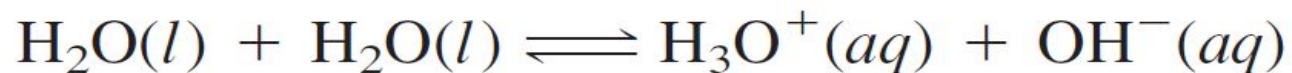


With each additional O atom, the Cl atom becomes effectively more electronegative (see Figure 16.7). As a result, the H atom becomes more acidic. The acid strengths increase in the following order:



Self-Ionization of Water

- Pure water is often considered a nonelectrolyte, precise measurements do show a very small conduction.
- This conduction results from self-ionization (or autoionization), a reaction in which two like molecules react to give ions.
- In the case of water, a proton from one H_2O molecule is transferred to another H_2O molecule, leaving behind an OH^- ion and forming a hydronium ion, H_3O^+ (aq).



$$K_c = \frac{[\text{H}_3\text{O}^+][\text{OH}^-]}{[\text{H}_2\text{O}]^2}$$

Because the concentration of ions formed is very small, the concentration of H_2O remains essentially constant

$$\underbrace{[\text{H}_2\text{O}]^2 K_c}_{\text{constant}} = [\text{H}_3\text{O}^+][\text{OH}^-]$$

We call the *equilibrium value of the ion product* $[\text{H}_3\text{O}^+][\text{OH}^-]$ *the ion-product constant for water*, which is written K_w . At 25°C, the value of K_w is 1.0×10^{-14} . Like any equilibrium constant, K_w varies with temperature. At body temperature (37°C), K_w equals 2.5×10^{-14} .

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] \\ = [\text{H}^+][\text{OH}^-]$$

Because we often write
 $\text{H}^+(\text{aq})$ for $\text{H}_3\text{O}^+(\text{aq})$

Using K_w , you can calculate the concentrations of $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ ions in pure water. These ions are produced in equal numbers in pure water, so their concentrations are equal. Let $X = [\text{H}_3\text{O}^+] = [\text{OH}^-]$. Then, substituting into the equation for the ion product constant,

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = X^2 \\ 1.0 \times 10^{-14} = X^2 \\ X = 1.0 \times 10^{-7}$$

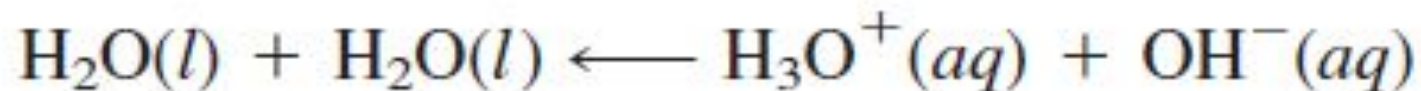
Solutions of a Strong Acid or Base

Consider an aqueous solution of a strong acid or base. Suppose you dissolve 0.10 mol HCl in 1.0 L of aqueous solution, giving 0.10M HCl. In addition of self-ionization , following reaction will also happen



As HCl is strong acid, 0.10 mol HCl in 1.0 L of solution will produce 0.10 mol $[\text{H}_3\text{O}^+]$, so the concentration of $[\text{H}_3\text{O}^+]$ ion from HCl is 0.10 M.

In pure water, the concentration of $[\text{H}_3\text{O}^+]$ produced is 1.0×10^{-7} M; in an acid solution, the contribution of $[\text{H}_3\text{O}^+]$ from water will be even smaller. Hence, according to Le Chatelier's by adding an acid, the self-ionization of water reverses until a new equilibrium is obtained.



Consequently, the concentration of $[\text{H}_3\text{O}^+]$ produced by the self-ionization of water ($1.0 \times 10^{-7} \text{ M}$) is negligible in comparison with that produced from HCl (0.10 M). So, 0.10 M HCl has a concentration of H_3O^+ ion equal to 0.10 M.

In a solution of a strong acid, you can normally ignore the self-ionization of water as a source of $[\text{H}_3\text{O}^+]$. The H_3O^+ concentration is usually determined by the strong acid concentration. Although you normally ignore the self-ionization of water in calculating the $[\text{H}_3\text{O}^+]$ concentration in a solution of a strong acid, the self-ionization equilibrium still exists and is responsible for a small concentration of OH^- ion.

$$\begin{aligned}K_w &= [\text{H}_3\text{O}^+][\text{OH}^-] \\1.0 \times 10^{-14} &= 0.10 \times [\text{OH}^-]\end{aligned}$$

Solving for $[\text{OH}^-]$,

$$[\text{OH}^-] = \frac{1.0 \times 10^{-14}}{0.10} = 1.0 \times 10^{-13}$$

The OH^- concentration is $1.0 \times 10^{-13} \text{ M}$.

Now consider a solution of a strong base, such as 0.010 M NaOH.

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$$
$$1.0 \times 10^{-14} = [\text{H}_3\text{O}^+] \times 0.010$$

Solving for H_3O^+ concentration,

$$[\text{H}_3\text{O}^+] = \frac{1.0 \times 10^{-14}}{0.010} = 1.0 \times 10^{-12}$$

The H_3O^+ concentration is $1.0 \times 10^{-12} \text{ M}$.

Calculating Concentrations of H_3O^+ and OH^- in Solutions of a Strong Acid or Base

Calculate the concentrations of hydronium ion and hydroxide ion at 25°C in:
a. 0.15 M HNO_3 , b. 0.010 M Ca(OH)_2 .

A strong acid (or base) is fully ionized in solution, so you can use its formula and molar concentration to determine the H_3O^+ (or OH^-) ion concentration. (Any contribution to this ion concentration from the self-ionization of water is negligible.) K_w relates the H_3O^+ and OH^- ion concentrations; therefore, you use this relation to obtain the OH^- concentration, given the H_3O^+ concentration, or vice versa.

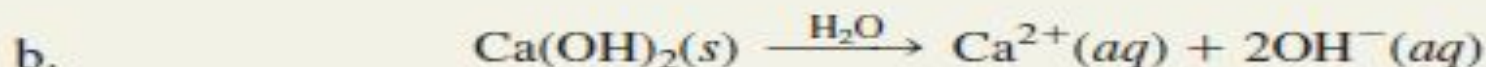
- a. Every mole of HNO_3 contributes one mole of H_3O^+ ion, so the H_3O^+ concentration is **0.15 M**. The OH^- concentration is obtained from the equation for K_w .

$$\begin{aligned}K_w &= [\text{H}_3\text{O}^+][\text{OH}^-] \\1.0 \times 10^{-14} &= 0.15 \times [\text{OH}^-]\end{aligned}$$

so

$$[\text{OH}^-] = \frac{1.0 \times 10^{-14}}{0.15} = 6.7 \times 10^{-14}$$

The concentration of OH^- in 0.15 M HNO_3 is **6.7×10^{-14} M**.



Every mole of Ca(OH)_2 that dissolves yields two moles of OH^- . Therefore, 0.010 M Ca(OH)_2 contains $2 \times 0.010 \text{ M OH}^- = \text{0.020 M OH}^-$. The H_3O^+ concentration is obtained from

$$\begin{aligned}K_w &= [\text{H}_3\text{O}^+][\text{OH}^-] \\1.0 \times 10^{-14} &= [\text{H}_3\text{O}^+] \times 0.020\end{aligned}$$

so

$$[\text{H}^+] = \frac{1.0 \times 10^{-14}}{0.020} = 5.0 \times 10^{-13}$$

The hydronium-ion concentration is **5.0×10^{-13} M**.

The hydrogen ion concentration of a fruit juice is 3.3×10^{-2} M. What is the pH of the juice ?
Is it acidic or basic ?

The definition of pH is

$$\text{pH} = -\log [\text{H}^+]$$

We are given $[\text{H}^+] = 3.3 \times 10^{-2}$

Substituting into the definition of pH, we get

$$\begin{aligned}\text{pH} &= -\log (3.3 \times 10^{-2}) \\ &= -(-1.48) \\ &= \mathbf{1.48}\end{aligned}$$

Since the pH is less than 7.00 the **solution is acidic.**

If a solution has a pH of 7.41, determine its H^+ concentration.

Since $\text{pH} = -\log [\text{H}^+]$

$$-\text{pH} = \log [\text{H}^+]$$

Now $[\text{H}^+]$ can be calculated by taking antilog of $-\text{pH}$:

$$[\text{H}^+] = \text{antilog} (-\text{pH})$$

In this case,

$$[\text{H}^+] = \text{antilog} (-7.41) = 10^{-7.41}$$

Therefore, the $[\text{H}^+]$ of the given solution is **$3.9 \times 10^{-8} \text{ M}$**

If a solution has a pH of 5.50 at 25°C, calculate its $[\text{OH}^-]$.

We know that :

$$\text{pH} + \text{pOH} = 14.00$$

$$\text{pOH} = 14.00 - \text{pH} = 14.00 - 5.50 = 8.50$$

Then

$$\text{pOH} = -\log [\text{OH}^-] = 8.50$$

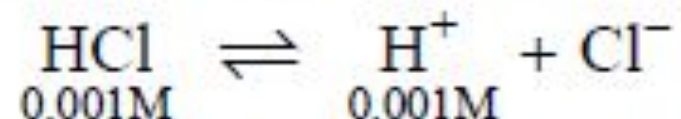
$$\log [\text{OH}^-] = -8.50$$

$$\begin{aligned} [\text{OH}^-] &= \text{antilog} (-8.50) \\ &= \mathbf{3.2 \times 10^{-9} \text{ M}} \end{aligned}$$



Calculate the pH of 0.001 M HCl.

HCl is a strong acid and it is completely dissociated in aqueous solution.



For every molecule of HCl, there is one H^+ . Therefore,

$$[\text{H}^+] = [\text{HCl}]$$

or

$$[\text{H}^+] = 0.001 \text{ M}$$

$\therefore$

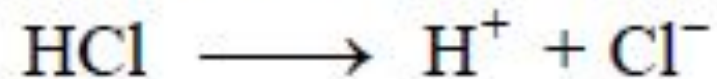
$$\begin{aligned}\text{pH} &= -\log(0.001) \\ &= -\log(1 \times 10^{-3}) \\ &= -\log 1 + 3 \log 10 \\ &= 0 + 3 = 3\end{aligned}$$

Therefore, the pH of 0.001 M HCl is **3**.

The pH of a solution of HCl is 2. Find out the amount of acid present in a litre of the solution.

$$\begin{aligned}\text{pH} &= 2 \\ -\log [\text{H}^+] &= 2\end{aligned}$$

The dissociation of HCl takes place according to the equation :



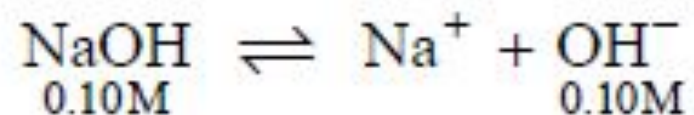
One molecule of HCl gives one ion of H^+ . Therefore,

$$[\text{H}^+] = [\text{HCl}] = 10^{-2} \text{ M}$$

$$\begin{aligned}\therefore \text{Amount of HCl in one litre} &= 10^{-2} \times \text{mol mass of HCl} \\ &= 10^{-2} \times 36.5 \\ &= \mathbf{0.365 \text{ g l}^{-1}}\end{aligned}$$

Determine the pH of 0.10 M NaOH solution.

NaOH is a strong base and it is completely dissociated in aqueous solution.



Therefore the concentration of OH^- ions is equal to that of the undissociated NaOH.

$$[\text{OH}^-] = [\text{NaOH}] = 0.10 \text{ M}$$

$[\text{H}^+]$ can be calculated by applying the expression by substituting the value of $K_w = 1 \times 10^{-14}$ and of $[\text{OH}^-]$

$$[\text{H}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{1 \times 10^{-14}}{0.10} = \frac{1 \times 10^{-14}}{10^{-1}} = 10^{-13} \text{ M}$$

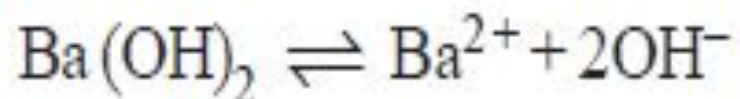
By definition

$$\begin{aligned} \text{pH} &= -\log [\text{H}^+] = -\log 10^{-13} \\ &= 13 \end{aligned}$$

Therefore the pH of 0.10 M NaOH is **13**.

Calculate the pH of a 0.020 M Ba(OH)₂ solution.

Barium hydroxide dissociates according to the equation



One molecule of barium hydroxide dissociates to give two OH⁻ ions. Therefore we have

$$[\text{OH}^-] = 0.040 \text{ M}$$

The pOH of the solution is, therefore,

$$\text{pOH} = -\log 0.040 = 1.40$$

We have that $\text{pH} + \text{pOH} = 14.00$

$$\therefore \text{pH} = 14.00 - 1.40 = 12.60$$

The pH of 0.020 M Ba(OH)₂ solution is **12.60**.

Do Your Self:

The pH of human arterial blood is 7.40. What is the hydronium-ion concentration?

$3.98 \times 10^{-8} \text{ M}$

A brand of carbonated beverage has a pH of 3.16. What is the hydronium-ion concentration of the beverage?

$6.92 \times 10^{-4} \text{ M}$

A 0.010 M solution of ammonia, NH_3 , has a pH of 10.6 at 25°C. What is the concentration of hydroxide ion?

$3.98 \times 10^{-4} \text{ M}$

Morphine is a narcotic that is used to relieve pain. A solution of morphine has a pH of 9.61 at 25°C. What is the hydroxide-ion concentration?

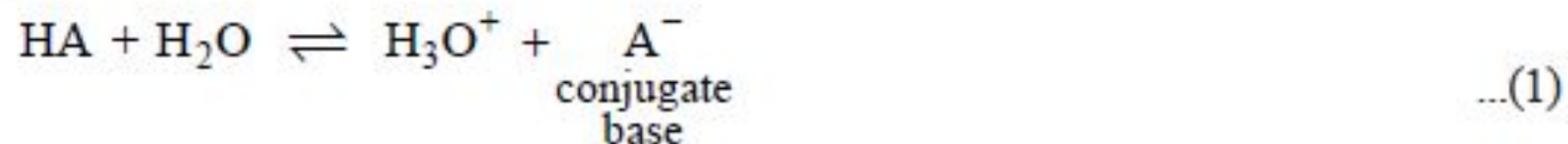
13.16

A 1.00-L aqueous solution contained 6.78 g of barium hydroxide, $\text{Ba}(\text{OH})_2$. What was the pH of the solution at 25°C?

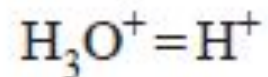
12.90

Relative Strength Of Acids

The strength of an acid depends on its ability to transfer its proton (H^+) to a base to form its conjugate base. When a monoprotic acid (HA) dissolves in water, it transfers its proton to water (a Bronsted base) to form hydronium ion (H_3O^+) and a conjugate base.



For simplifying our discussion, we take



Thus we can write the equilibrium reaction (1) as



This equation represents the dissociation of the acid HA into H^+ ion and A^- ion.

Applying the Law of Mass action to the acid dissociation equilibrium, we can write

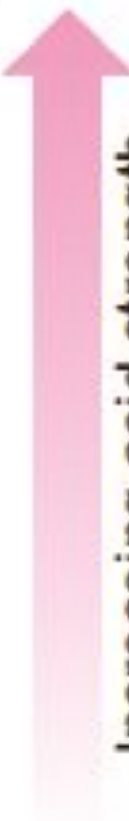
$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} \quad \dots(3)$$

where K_a is called the acid dissociation constant. In dilute solution of the acid (HA) we note that the concentration of liquid water remains essentially constant. Therefore, the terms included in the equilibrium expression (3).

The strength of an acid is defined as the concentration of H^+ ions in its aqueous solution at a given temperature.

From the equilibrium (3), it is evident that the concentration of H^+ ions, $[H^+]$, depends on the value of K_a . **Therefore, the value of K_a for a particular acid is a measure of its acid strength or acidity.**

Formula	Name	K_a
HSO_4^-	Hydrogen sulphate Ion	1.2×10^{-2}
HClO_2	Chlorous acid	1.2×10^{-2}
ClCH_2COOH	Monochloroacetic acid	1.35×10^{-3}
HF	Hydrofluoric acid	7.2×10^{-4}
HNO_2	Nitrous acid	4.0×10^{-4}
CH_3COOH	Acetic acid	1.8×10^{-5}
HOCl	Hypochlorous acid	3.5×10^{-8}
HCN	Hydrocyanic acid	6.2×10^{-10}
NH_4^+	Ammonium ion	5.6×10^{-10}
HOC_6H_5	Phenol	1.6×10^{-10}



Increasing acid strength

Calculation of Relative strength of Weak acids from K_a

We have seen that for an acid in aqueous solution we have, $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$

and

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

Let C moles per litre be the concentration of the acid and α its degree of dissociation. Then,

$$[\text{H}^+] = C\alpha$$

$$[\text{A}^-] = C\alpha$$

$$[\text{HA}] = C(1 - \alpha)$$

Substituting the values in the equilibrium expression we have

$$\begin{aligned} K_a &= \frac{C\alpha \times C\alpha}{C(1 - \alpha)} \\ &= \frac{C\alpha^2}{(1 - \alpha)} \end{aligned}$$

For weak acids $1 - \alpha \approx 1$. Therefore,

$$K_a = C\alpha^2$$

For two different acids, 1 and 2, let the degree of dissociation be α_1 and α_2 ; and the dissociation constants K_1 and K_2 . Then,

$$\text{for acid 1} \quad K_1 = C\alpha_1^2 \quad \dots(1)$$

$$\text{for acid 2} \quad K_2 = C\alpha_2^2 \quad \dots(2)$$

Dividing equation (1) by (2), we get,

$$\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_1}{K_2}}$$

Since $[H^+]$ is a measure of acid strength and it depends on the degree of dissociation α , we can write

$$\frac{\text{Strength of acid 1}}{\text{Strength of acid 2}} = \sqrt{\frac{K_1}{K_2}}$$

The dissociation constants of formic acid and acetic acid are 21.4×10^{-5} and 1.81×10^{-5} respectively. Find the relative strengths of the acids.

Calculating the pH of Weak Acid

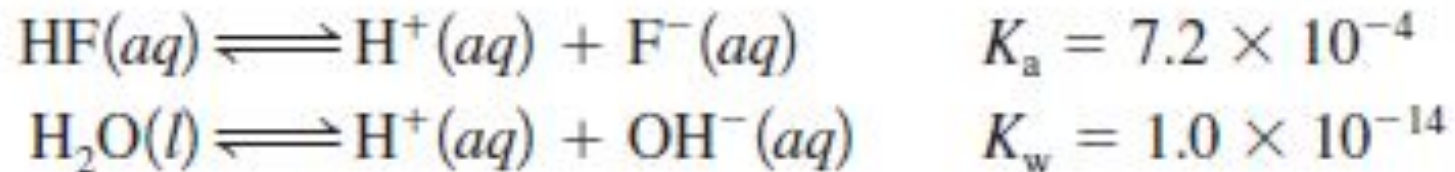
Since a weak acid dissolved in water can be viewed as a prototype of almost any Equilibrium occurring in aqueous solution, we will proceed carefully and systematically. Although some of the procedures we develop here may seem unnecessary, they will become essential as the problems become more complicated. We will develop the necessary strategies by calculating the pH of a 1.00-*M* solution of HF ($K_a = 7.2 \times 10^{-4}$).

Step-1: finding the major species in the solution.

From its small K_a value, we know that hydrofluoric acid is a weak acid and will be dissociated only to a slight extent. Thus when we write the major species, the hydrofluoric acid will be represented in its dominant form, as HF. The major species in solution are HF and H_2O .

Step-3: writing the possible equilibrium equation

The next step (since this is a pH problem) is to decide which of the major species can furnish H^+ ions. Actually, both major species can do so:



Step-3: determining the dominant acid

By comparing K_a for HF with K_w for H_2O , we see that hydrofluoric acid, although weak, is still a much stronger acid than water. Thus we will assume that hydrofluoric acid will be the dominant source of H^+ . We will ignore the tiny contribution by water.

Step-4: writing the equilibrium equation

Therefore, it is the dissociation of HF that will determine the equilibrium concentration of H^+ and hence the pH:



The equilibrium expression is

$$K_a = 7.2 \times 10^{-4} = \frac{[H^+][F^-]}{[HF]}$$

Step-5: writing the initial concentration

First, we list the initial concentrations, the concentrations *before the reaction of interest has proceeded to equilibrium*. Before any HF dissociates, the concentrations of the species in the equilibrium are

$$[HF]_0 = 1.00\ M \quad [F^-]_0 = 0 \quad [H^+]_0 = 10^{-7}\ M \approx 0$$

(Note that the zero value for $[\text{H}^+]_0$ is an approximation, since we are neglecting the H^+ ions from the autoionization of water.)

Step-4: writing the equilibrium equation

The next step is to determine the change required to reach equilibrium. Since some HF will dissociate to come to equilibrium (but this amount is presently unknown), we let x be the change in the concentration of HF that is required to achieve equilibrium. That is, we assume that x mol/L HF will dissociate to produce x mol/L H^+ and x mol/L F^- as the system adjusts to its equilibrium position. Now the equilibrium concentrations can be defined in terms of x :

$$[\text{HF}] = [\text{HF}]_0 - x = 1.00 - x$$

$$[\text{F}^-] = [\text{F}^-]_0 + x = 0 + x = x$$

$$[\text{H}^+] = [\text{H}^+]_0 + x \approx 0 + x = x$$

Substituting these equilibrium concentrations into the equilibrium expression gives

$$K_a = 7.2 \times 10^{-4} = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]} = \frac{(x)(x)}{1.00 - x}$$

If x is very small compared to 1.00, the term in the denominator can be approximated as follows:

$$1.00 - x \approx 1.00$$

The equilibrium expression then becomes

$$7.2 \times 10^{-4} = \frac{(x)(x)}{1.00 - x} \approx \frac{(x)(x)}{1.00}$$

which yields

$$x^2 \approx (7.2 \times 10^{-4})(1.00) = 7.2 \times 10^{-4}$$

$$x \approx \sqrt{7.2 \times 10^{-4}} = 2.7 \times 10^{-2}$$

How valid is the approximation that $[\text{HF}] \approx 1.00 \text{ M}$? Because this question will arise often in connection with acid–base equilibrium calculations, we will consider it carefully. The validity of the approximation depends on how much accuracy we demand for the calculated value of $[\text{H}^+]$. Typically, the K_a values for acids are known to an accuracy of only about $\pm 5\%$. It is reasonable therefore to apply this figure when determining the validity of the approximation

$$[\text{HA}]_0 - x \approx [\text{HA}]_0$$

We will use the following test. First, we calculate the value of x by making the approximation

$$K_a = \frac{x^2}{[\text{HA}]_0 - x} \approx \frac{x^2}{[\text{HA}]_0}$$

where

$$x^2 \approx K_a[\text{HA}]_0 \quad \text{and} \quad x \approx \sqrt{K_a[\text{HA}]_0}$$

We then compare the sizes of x and $[\text{HA}]_0$. If the expression

$$\frac{x}{[\text{HA}]_0} \times 100\%$$

is less than or equal to 5%, the value of x is small enough that the approximation

$$[\text{HA}]_0 - x \approx [\text{HA}]_0$$

will be considered valid.

In our example,

$$x = 2.7 \times 10^{-2} \text{ mol/L}$$

$$[\text{HA}]_0 = [\text{HF}]_0 = 1.00 \text{ mol/L}$$

and

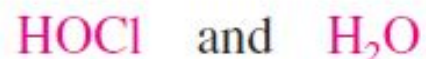
$$\frac{x}{[\text{HA}]_0} \times 100 = \frac{2.7 \times 10^{-2}}{1.00} \times 100\% = 2.7\%$$

The approximation we made is considered valid, and the value of x calculated using that approximation is acceptable. Thus

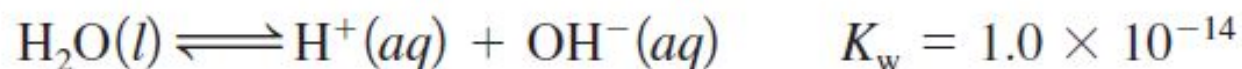
$$x = [\text{H}^+] = 2.7 \times 10^{-2} \text{ M} \quad \text{and} \quad \text{pH} = -\log(2.7 \times 10^{-2}) = 1.57$$

The hypochlorite ion (OCl^-) is a strong oxidizing agent often found in household bleaches and disinfectants. It is also the active ingredient that forms when swimming-pool water is treated with chlorine. In addition to its oxidizing abilities, the hypochlorite ion has a relatively high affinity for protons (it is a much stronger base than Cl^- , for example) and forms the weakly acidic hypochlorous acid (HOCl , $K_a = 3.5 \times 10^{-8}$). Calculate the pH of a 0.100- M aqueous solution of hypochlorous acid.

1. We list the major species. Since HOCl is a weak acid and remains mostly undissociated, the major species in a 0.100- M HOCl solution are



2. Both species can produce H^+ :



3. Since HOCl is a significantly stronger acid than H_2O , it will dominate in the production of H^+ .
4. We therefore use the following equilibrium expression:

$$K_a = 3.5 \times 10^{-8} = \frac{[\text{H}^+][\text{OCl}^-]}{[\text{HOCl}]}$$

5. The initial concentrations appropriate for this equilibrium are

$$[\text{HOCl}]_0 = 0.100 \text{ M}$$

$$[\text{OCl}^-]_0 = 0$$

$$[\text{H}^+]_0 \approx 0 \quad (\text{We neglect the contribution from H}_2\text{O.})$$

6. Since the system will reach equilibrium by the dissociation of HOCl, let x be the amount of HOCl (in mol/L) that dissociates in reaching equilibrium.

7. The equilibrium concentrations in terms of x are

$$[\text{HOCl}] = [\text{HOCl}]_0 - x = 0.100 - x$$

$$[\text{OCl}^-] = [\text{OCl}^-]_0 + x = 0 + x = x$$

$$[\text{H}^+] = [\text{H}^+]_0 + x \approx 0 + x = x$$

8. Substituting these concentrations into the equilibrium expression gives

$$K_a = 3.5 \times 10^{-8} = \frac{(x)(x)}{0.100 - x}$$

9. Since K_a is so small, we can expect a small value for x . Thus we make the approximation $[HA]_0 - x \approx [HA]_0$, or $0.100 - x \approx 0.100$, which leads to the expression

$$K_a = 3.5 \times 10^{-8} = \frac{x^2}{0.100 - x} \approx \frac{x^2}{0.100}$$

Solving for x gives

$$x = 5.9 \times 10^{-5}$$

10. The approximation $0.100 - x \approx 0.100$ must be validated. To do this, we compare x to $[HOCl]_0$:

$$\frac{x}{[HA]_0} \times 100 = \frac{x}{[HOCl]_0} \times 100 = \frac{5.9 \times 10^{-5}}{0.100} \times 100 = 0.059\%$$

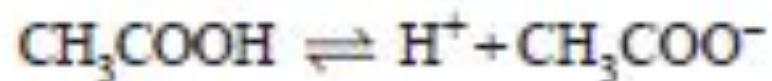
Since this value is much less than 5%, the approximation is considered valid.

11. We calculate $[H^+]$ and pH:

$$[H^+] = x = 5.9 \times 10^{-5} M \quad \text{and} \quad \text{pH} = 4.23$$

Calculate the pH of 0.1 M CH_3COOH . The dissociation constant of acetic acid is 1.8×10^{-5} .

Acetic acid dissociates in aqueous solution as



The K_a for acetic acid is stated as

$$K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 1.8 \times 10^{-5} \quad \dots(1)$$

One molecule of CH_3COOH dissociates to form one H^+ and one CH_3COO^- . Thus,

$$[\text{H}^+] = [\text{CH}_3\text{COO}^-]$$

If x be the concentration of H^+ ion at equilibrium, the various concentrations at the equilibrium may be written as

$$[\text{CH}_3\text{COOH}] = 0.10 - x$$

$$[\text{H}^+] = x$$

$$[\text{CH}_3\text{COO}^-] = x$$

Since the value of x is negligible relative to 0.10, $(0.10 - x)$ is approximately equal to 0.10. Substituting the values in (1), we have

$$K_a = \frac{x \times x}{0.10} = 1.8 \times 10^{-5}$$

Solving for x ,

$$\frac{x^2}{0.10} = 1.8 \times 10^{-5}$$

or

$$x^2 = (1.0 \times 10^{-1})(1.8 \times 10^{-5}) = 1.8 \times 10^{-6}$$

$$x = \sqrt{1.8 \times 10^{-6}} = 1.34 \times 10^{-3} \text{ M}$$

Thus the hydrogen concentration $[H^+]$ of 0.1 M CH_3COOH is $1.34 \times 10^{-3} \text{ M}$

$$\begin{aligned} \text{pH} &= -\log [H^+] = -\log (1.34 \times 10^{-3}) \\ &= +3 - \log 1.34 \\ &= 2.87 \end{aligned}$$

Thus the pH of 0.1 M CH_3COOH solution is 2.87.

The pH of Weak Acid Mixtures

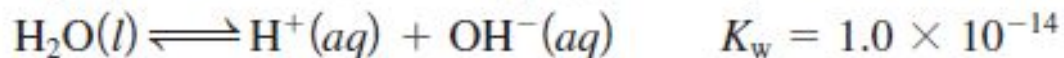
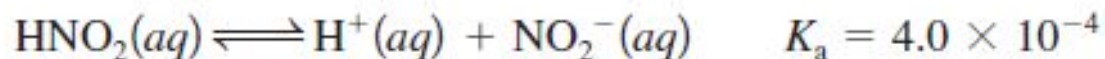
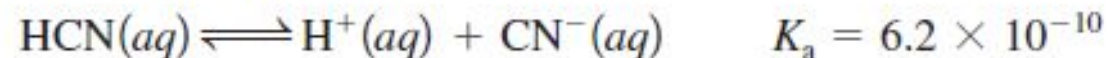
Calculate the pH of a solution that contains 1.00 *M* HCN ($K_a = 6.2 \times 10^{-10}$) and 5.00 *M* HNO₂ ($K_a = 4.0 \times 10^{-4}$). Also calculate the concentration of cyanide ion (CN⁻) in this solution at equilibrium.

Solution

Since HCN and HNO₂ are both weak acids and are largely undissociated, the major species in the solution are



All three of these components produce H⁺:



A mixture of three acids might lead to a very complicated problem. However, the situation is greatly simplified by the fact that even though HNO₂ is a weak acid, it is much stronger than the other two acids present (as revealed by the *K* values). Thus HNO₂ can be assumed to be the dominant producer of H⁺, and we will focus on the equilibrium expression

$$K_a = 4.0 \times 10^{-4} = \frac{[\text{H}^+][\text{NO}_2^-]}{[\text{HNO}_2]}$$

The initial concentrations, the definition of x , and the equilibrium concentrations are as follows:

Initial Concentration (mol/L)		Equilibrium Concentration (mol/L)
$[\text{HNO}_2]_0 = 5.00$ $[\text{NO}_2^-]_0 = 0$ $[\text{H}^+]_0 \approx 0$	$\xrightarrow[x \text{ mol/L HNO}_2 \text{ dissociates}]{} \rightarrow$	$[\text{HNO}_2] = 5.00 - x$ $[\text{NO}_2^-] = x$ $[\text{H}^+] = x$

It is convenient to represent these concentrations in the following shorthand form (called an ICE table):

	$\text{HNO}_2(aq)$	$\rightleftharpoons$	$\text{H}^+(aq)$	+	$\text{NO}_2^-(aq)$
Initial	5.00		0		0
Change	$-x$		$+x$		$+x$
Equilibrium	$5.00 - x$		x		x

Substituting the equilibrium concentrations in the equilibrium expression and making the approximation that $5.00 - x = 5.00$ give

$$K_a = 4.0 \times 10^{-4} = \frac{(x)(x)}{5.00 - x} \approx \frac{x^2}{5.00}$$

We solve for x :

$$x = 4.5 \times 10^{-2}$$

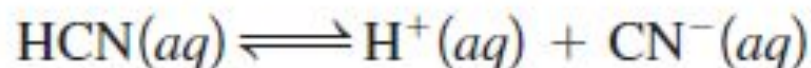
Using the 5% rule, we show that the approximation is valid:

$$\frac{x}{[\text{HNO}_2]_0} \times 100 = \frac{4.5 \times 10^{-2}}{5.00} \times 100 = 0.90\%$$

Therefore,

$$[\text{H}^+] = x = 4.5 \times 10^{-2} M \quad \text{and} \quad \text{pH} = 1.35$$

We also want to calculate the equilibrium concentration of cyanide ion in this solution. The CN^- ions in this solution come from the dissociation of HCN:



Although the position of this equilibrium lies far to the left and does not contribute *significantly* to $[\text{H}^+]$, HCN is the *only source* of CN^- . Thus we must consider the extent of the dissociation of HCN to calculate $[\text{CN}^-]$. The equilibrium expression for the preceding reaction is

$$K_a = 6.2 \times 10^{-10} = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]}$$

We know $[H^+]$ for this solution from the results of the first part of the problem. It is important to understand that *there is only one kind of H^+ in this solution*. It does not matter from which acid the H^+ ions originate. The equilibrium $[H^+]$ we need to insert into the HCN equilibrium expression is $4.5 \times 10^{-2} M$, even though the H^+ was contributed almost entirely from the dissociation of HNO_2 . What is $[HCN]$ at equilibrium? We know $[HCN]_0 = 1.00 M$, and since K_a for HCN is so small, a negligible amount of HCN will dissociate. Thus

$$[HCN] = [HCN]_0 - \text{amount of HCN dissociated} \approx [HCN]_0 = 1.00 M$$

Since $[H^+]$ and $[HCN]$ are known, we can find $[CN^-]$ from the equilibrium expression:

$$K_a = 6.2 \times 10^{-10} = \frac{[H^+][CN^-]}{[HCN]} = \frac{(4.5 \times 10^{-2})[CN^-]}{1.00}$$
$$[CN^-] = \frac{(6.2 \times 10^{-10})(1.00)}{4.5 \times 10^{-2}} = 1.4 \times 10^{-8} M$$

Note the significance of this result. Since $[\text{CN}^-] = 1.4 \times 10^{-8} \text{ M}$ and HCN is the only source of CN^- , this means that only $1.4 \times 10^{-8} \text{ mol/L}$ of HCN dissociated. This is a very small amount compared with the initial concentration of HCN, which is exactly what we would expect from its very small K_a value, and $[\text{HCN}] = 1.00 \text{ M}$ as assumed.

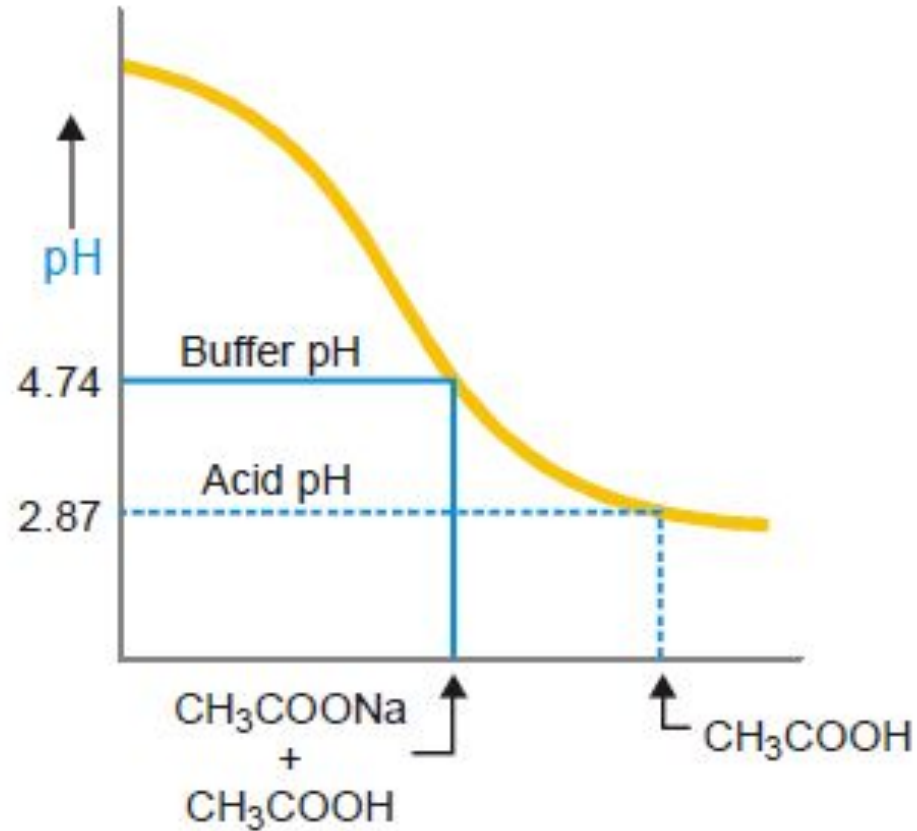
What Is A Buffer Solution ?

A buffer solution is one which maintains its pH fairly constant even upon the addition of small amounts of acid or base.

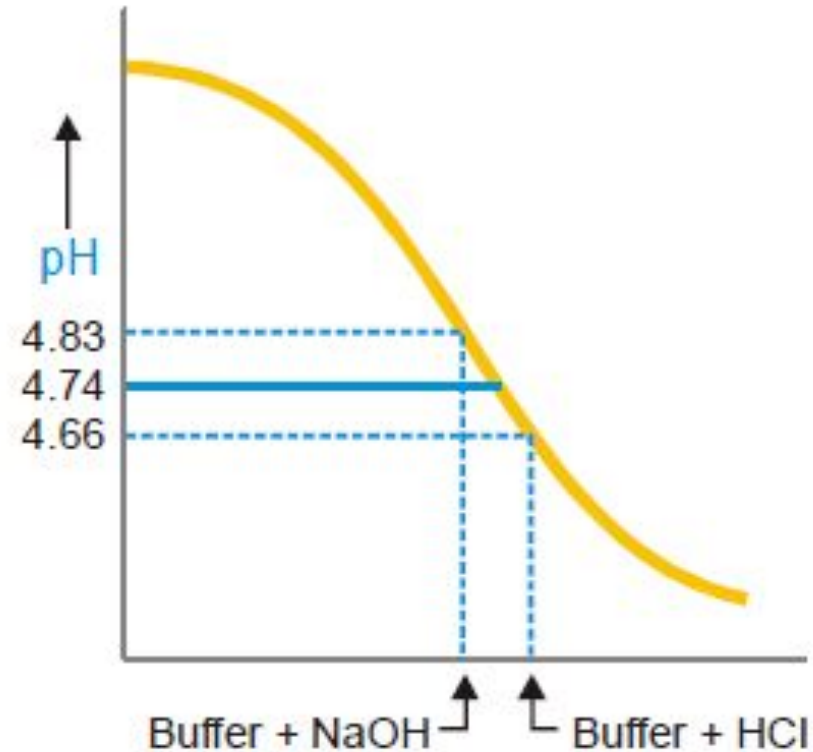
Two common types of buffer solutions are :

(1) a weak acid together with a salt of the same acid with a strong base. These are called Acid buffers e.g., $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$.

(2) a weak base and its salt with a strong acid. These are called Basic buffers. e.g., $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$.



■ **Figure 27.11**
The buffer solution ($\text{CH}_3\text{COOH}/\text{CH}_3\text{COONa}$) has a higher pH than the acid itself.



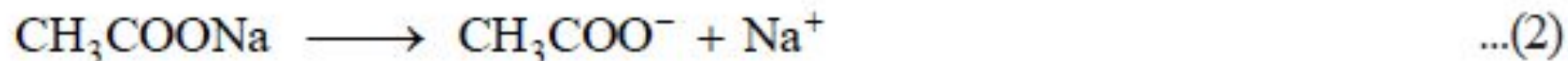
■ **Figure 27.12**
The pH of buffer changes only slightly upon addition of an acid (0.01 moles) or base (0.01 mole NaOH).

How A Buffer Operates ?

The pH of the buffer is governed by the equilibrium

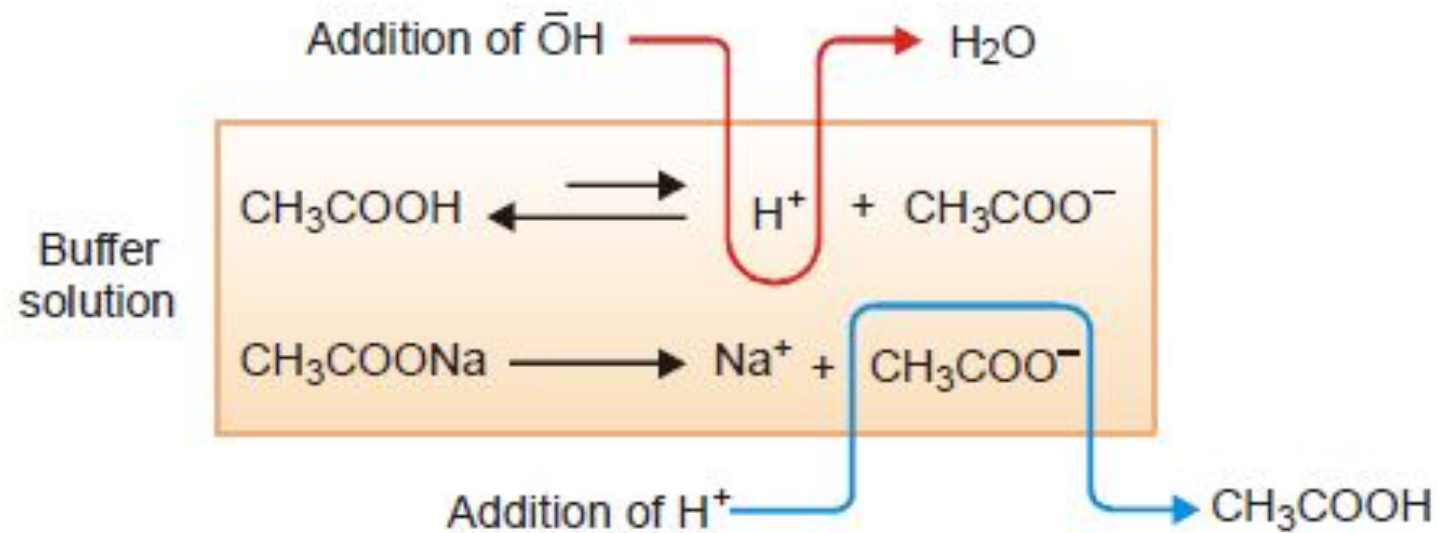


The buffer solution has a large excess of CH_3COO^- ions produced by complete ionisation of sodium acetate,

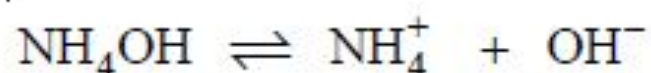


Addition of HCl. Upon the addition of HCl, the increase of H^+ ions is counteracted by association with the excess of acetate ions to form unionised CH_3COOH . Thus the added H^+ ions are neutralised and the pH of the buffer solution remains virtually unchanged. However owing to the increased concentration of CH_3COOH , the equilibrium (1) shifts slightly to the right to increase H^+ ions. This explains the marginal decrease of pH of the buffer solution on addition of HCl.

(2) Addition of NaOH. When NaOH is added to the buffer solution, the additional OH^- ions combine with H^+ ions of the buffer to form water molecules. As a result the equilibrium (1) shifts to the right to produce more and more H^+ ions till practically all the excess OH^- ions are neutralised and the original buffer pH restored. However, a new equilibrium system is set up in which $[\text{CH}_3\text{COOH}]$ is lower than it was in the original buffer. Consequently $[\text{H}^+]$ is also slightly less and pH slightly higher than the buffer pH values

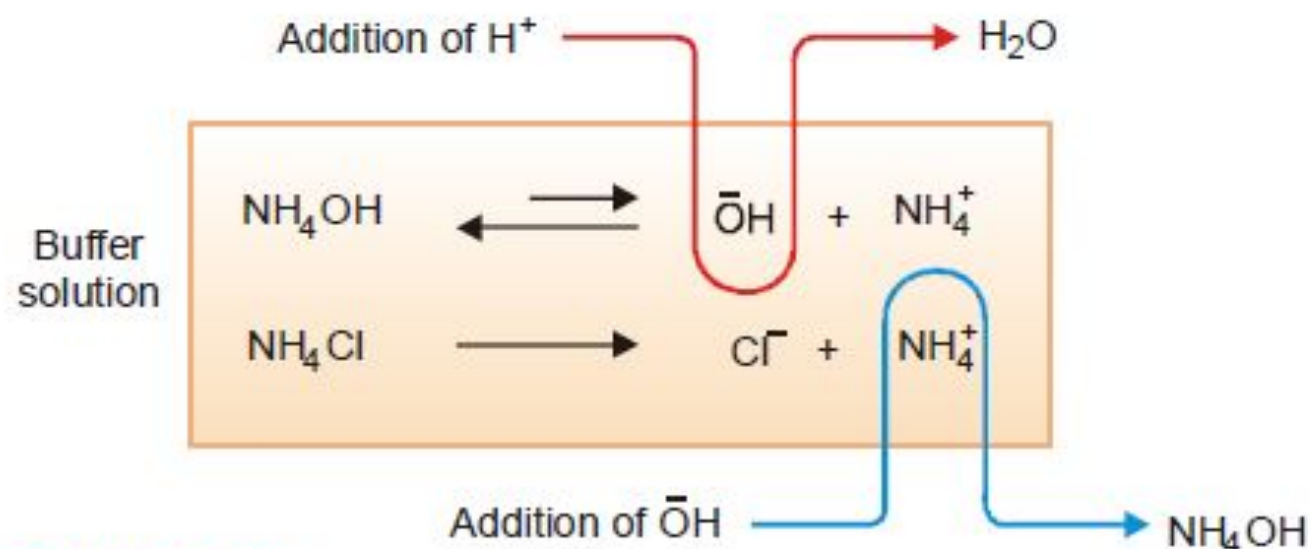


Operation of a Basic buffer as $\text{NH}_4\text{OH}/\text{NH}_4\text{Cl}$ can also be explained on the same lines as of an acid buffer (Fig. 27.14) upon addition of HCl the H^+ ions combine with OH^- ions of the buffer to form water molecules. The equilibrium,



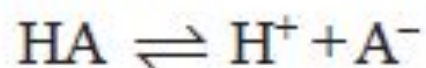
is shifted to the right till all the additional H^+ ions are neutralised and the original buffer pH restored.

When NaOH is added to the buffer solution, OH^- ions associate with excess of NH_4^+ ions to form unassociated NH_4OH . Thus the pH of the buffer is maintained approximately constant.



The pH of an acid buffer can be calculated from the dissociation constant, K_a , of the weak acid and the concentrations of the acid and the salt used.

The dissociation expression of the weak acid, HA, may be represented as



and

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

or

$$[\text{H}^+] = K_a \times \frac{[\text{HA}]}{[\text{A}^-]} \quad \dots(1)$$

The weak acid is only slightly dissociated and its dissociation is further depressed by the addition of the salt (Na^+A^-) which provides A^- ions (Common ion effect). As a result the equilibrium concentration of the unionised acid is nearly equal to the initial concentration of the acid. The equilibrium concentration $[\text{A}^-]$ is presumed to be equal to the initial concentration of the salt added since it is completely dissociated. Thus we can write the equation (1) as

$$[\text{H}^+] = K_a \times \frac{[\text{acid}]}{[\text{salt}]} \quad (2)$$

where [acid] is the initial concentration of the added acid and [salt] that of the salt used.

Taking negative logs of both sides of the equation (2), we have

$$-\log[H^+] = -\log K_a - \log \frac{[\text{acid}]}{[\text{salt}]} \quad \dots(3)$$

But $-\log[H^+] = \text{pH}$ and $-\log K_a = \text{p}K_a$

Thus from (3) we have

$$\text{pH} = \text{p}K_a - \log \frac{[\text{acid}]}{[\text{salt}]} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Hence
$$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

This relationship is called the **Henderson-Hasselbalch equation** or simply **Henderson equation**.

In a similar way, the Henderson-Hasselbalch equation for a basic buffer can be derived. This can be stated as :

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

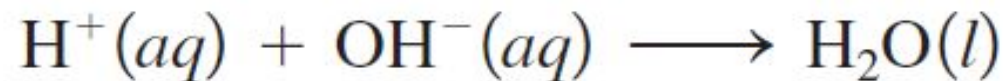
Titration and pH Curves

Titration is commonly used to determine the amount of acid or base in a solution. This process involves a solution of known concentration (the titrant) delivered from a burette into the unknown solution until the substance being analyzed is just consumed. **The stoichiometric (equivalence) point is often signaled by the color change of an indicator.**

The progress of an acid–base titration is often monitored by plotting the pH of the solution being analyzed as a function of the amount of titrant added. Such a plot is called a pH curve or titration curve.

Strong Acid–Strong Base Titrations

The net ionic reaction for a strong acid–strong base titration is



To compute $[\text{H}^+]$ at a given point in the titration, we must determine the amount of H^+ that remains at that point and divide by the total volume of the solution. Before we proceed, we need to consider a new unit, which is especially convenient for titrations. Since titrations usually involve small quantities (burets are typically graduated in milliliters), the mole is inconveniently large. Therefore, we will use the **millimole** (abbreviated **mmol**), which, as the prefix indicates, is a thousandth of a mole:

$$1 \text{ mmol} = \frac{1 \text{ mol}}{1000} = 10^{-3} \text{ mol}$$

So far we have defined molarity only in terms of moles per liter. We can now define it in terms of *millimoles per milliliter*, as shown below:

$$\text{Molarity} = \frac{\text{mol solute}}{\text{L solution}} = \frac{\frac{\text{mol solute}}{1000}}{\frac{\text{L solution}}{1000}} = \frac{\text{mmol solute}}{\text{mL solution}}$$

A 1.0-M solution thus contains 1.0 mole of solute per liter of solution or, equivalently, 1.0 mmol of solute per milliliter of solution. Just as we obtain the number of moles of solute from the product of the volume in liters and the molarity, we obtain the number of millimoles of solute from the product of the volume in milliliters and the molarity:

$$\text{Number of mmol} = \text{volume (in mL)} \times \text{molarity}$$

CASE STUDY Strong Acid–Strong Base Titration

We will illustrate the calculations involved in a strong acid–strong base titration by considering the titration of 50.0 mL of 0.200 M HNO₃ with 0.100 M NaOH. We will calculate the pH of the solution at selected points during the course of the titration, where specific volumes of 0.100 M NaOH have been added

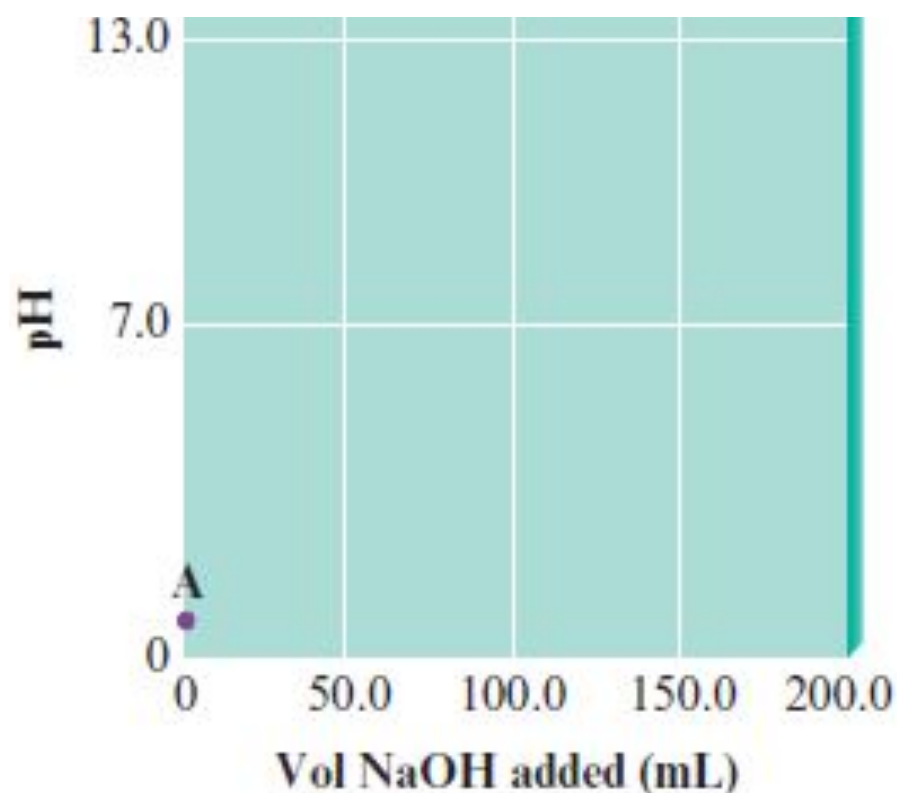
A. No NaOH has been added.

Since HNO_3 is a strong acid (is completely dissociated), the solution contains the major species



and the pH is determined by the H^+ from the nitric acid. Since 0.200 M HNO_3 contains 0.200 M H^+ ,

$$[\text{H}^+] = 0.200\text{ M} \quad \text{and} \quad \text{pH} = 0.699$$



B. 10.0 mL of 0.100 M NaOH has been added.

In the mixed solution *before any reaction occurs*, the major species are



Note that large quantities of both H^+ and OH^- are present. The 1.00 mmol (10.0 mL $\times$ 0.100 M) of added OH^- will react with 1.00 mmol H^+ to form water:

	H^+	+	OH^-	$\longrightarrow$	H_2O
Before reaction	50.0 mL $\times$ 0.200 M = 10.0 mmol		10.0 mL $\times$ 0.100 M = 1.00 mmol		
After reaction	10.0 – 1.00 = 9.0 mmol		1.00 – 1.00 = 0		

and the pH will be determined by the H^+ remaining:

$$[\text{H}^+] = \frac{\text{mmol H}^+ \text{ left}}{\text{volume of solution (mL)}} = \frac{9.0 \text{ mmol}}{(50.0 + 10.0) \text{ mL}} = 0.15 \text{ M}$$

Original volume of
 HNO_3 solution

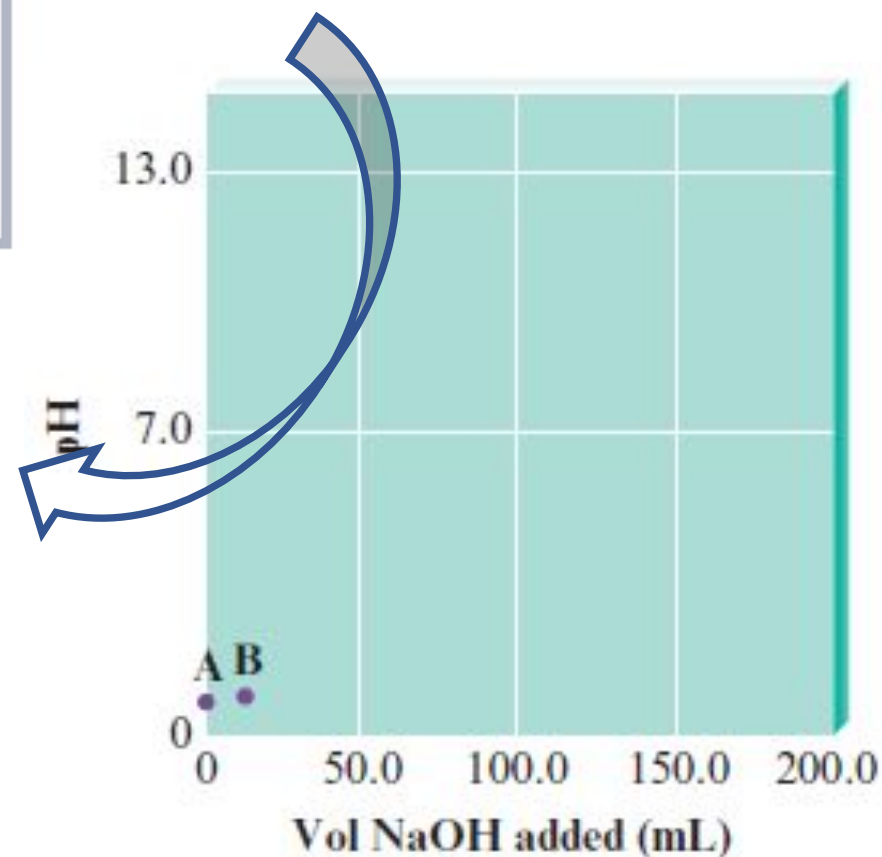
Volume of
NaOH added

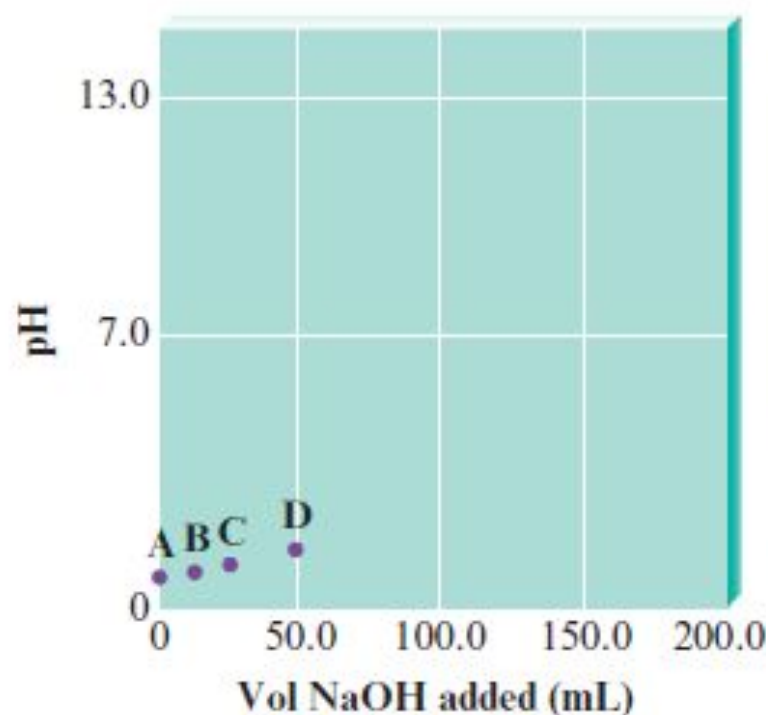
$$\text{pH} = -\log(0.15) = 0.82$$

After the reaction, the solution contains



(the OH^- ions have been consumed)





C. 20.0 mL (total) of 0.100 M NaOH has been added.

We consider this point from the perspective that a total of 20.0 mL NaOH has been added to the *original* solution, rather than that 10.0 mL has been added to the solution from point B. It is best to go back to the original solution each time so that a mistake made at an earlier point does not show up in each succeeding calculation. As before, the added OH^- will react with H^+ to form water:

	H^+	+	OH^-	$\longrightarrow$	H_2O
Before reaction	$50.0 \text{ mL} \times 0.200 \text{ M}$ $= 10.0 \text{ mmol}$		$20.0 \text{ mL} \times 0.100 \text{ M}$ $= 2.00 \text{ mmol}$		
After reaction	$10.0 - 2.00$ $= 8.00 \text{ mmol}$		$2.00 - 2.00$ $= 0 \text{ mmol}$		

After the reaction

$$[\text{H}^+] = \frac{8.00 \text{ mmol}}{(50.0 + 20.0) \text{ mL}} = 0.11 \text{ M}$$

pH = 0.942

D. 50.0 mL (total) of 0.100 M NaOH has been added.
 Proceeding exactly as for points B and C, the pH is found to be 1.301.

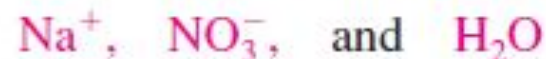
E. 100.0 mL (total) of 0.100 M NaOH has been added.
 this point the amount of NaOH that has been added is

$$100.0 \text{ mL} \times 0.100 \text{ M} = 10.0 \text{ mmol}$$

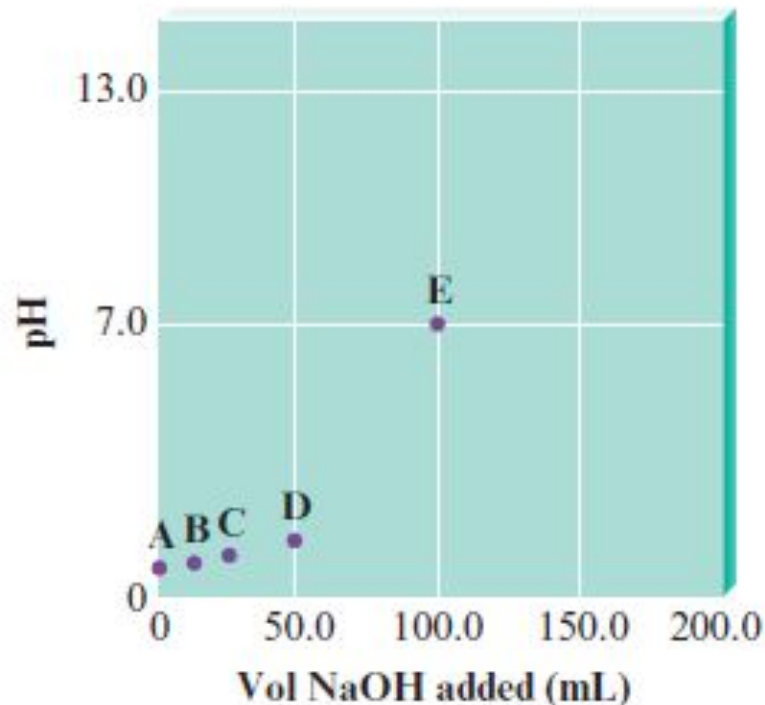
the original amount of nitric acid was

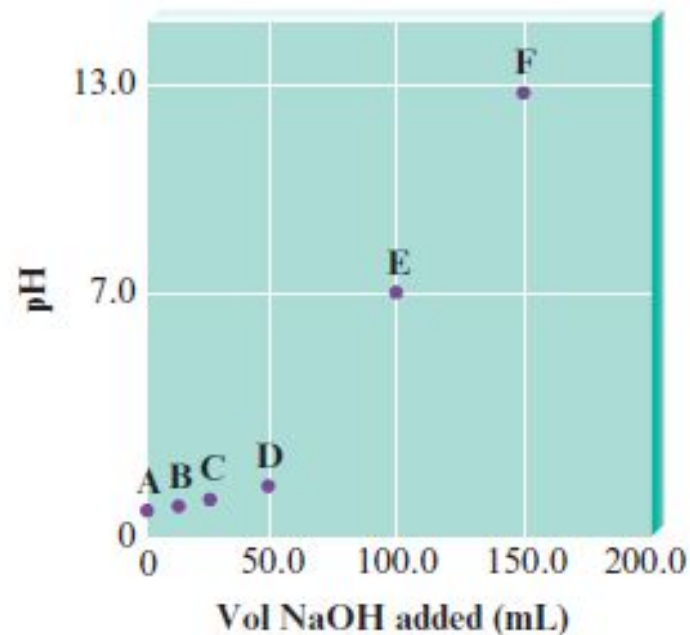
$$50.0 \text{ mL} \times 0.200 \text{ M} = 10.0 \text{ mmol}$$

Enough OH^- has been added to react exactly with the H^+ from the nitric acid. This is the **stoichiometric point**, or **equivalence point**, of the titration. At this point the major species in solution are



Since Na^+ has no acid or base properties and NO_3^- is the anion of the strong acid HNO_3 and is therefore a very weak base, neither NO_3^- nor Na^+ affects the pH, and the solution is neutral (the pH is 7.00).





F. 150.0 mL (total) of 0.100 M NaOH has been added.
The stoichiometric calculations for the titration reaction are as follows:

	H^+	+	OH^-	$\longrightarrow$	H_2O
Before reaction	$50.0 \text{ mL} \times 0.200 \text{ M}$ $= 10.0 \text{ mmol}$		$150.0 \text{ mL} \times 0.100 \text{ M}$ $= 15.0 \text{ mmol}$		
After reaction	$10.0 - 10.0$ $= 0 \text{ mmol}$		$15.0 - 10.0$ $= 5.0 \text{ mmol}$		
			$\uparrow$ Excess OH^- added		

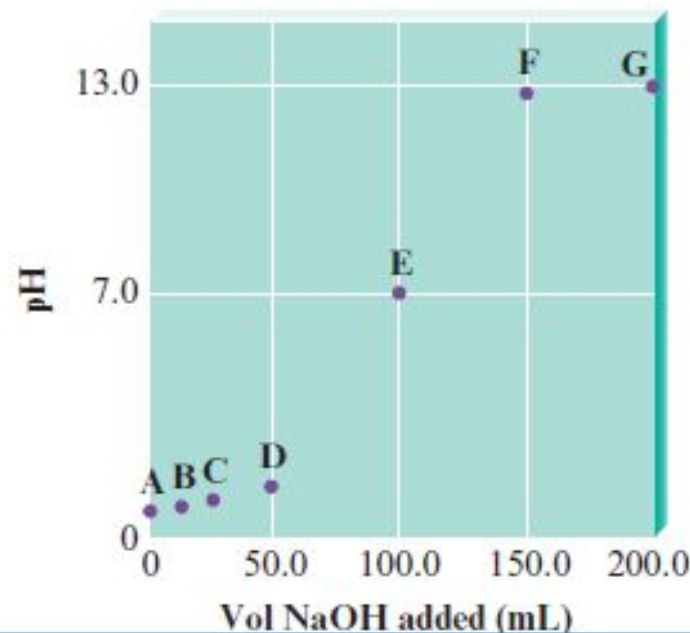
Now OH^- is *in excess* and will determine the pH:

$$[\text{OH}^-] = \frac{\text{mmol OH}^- \text{ in excess}}{\text{volume (mL)}} = \frac{5.0 \text{ mmol}}{(50.0 + 150.0) \text{ mL}} = \frac{5.0 \text{ mmol}}{200.0 \text{ mL}} = 0.025 \text{ M}$$

Since $[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$,

$$[\text{H}^+] = \frac{1.0 \times 10^{-14}}{2.5 \times 10^{-2}} = 4.0 \times 10^{-13} \text{ M} \quad \text{and} \quad \text{pH} = 12.40$$

G. 200.0 mL (total) of 0.100 M NaOH has been added.
Proceeding as for point F, the pH is found to be 12.60.

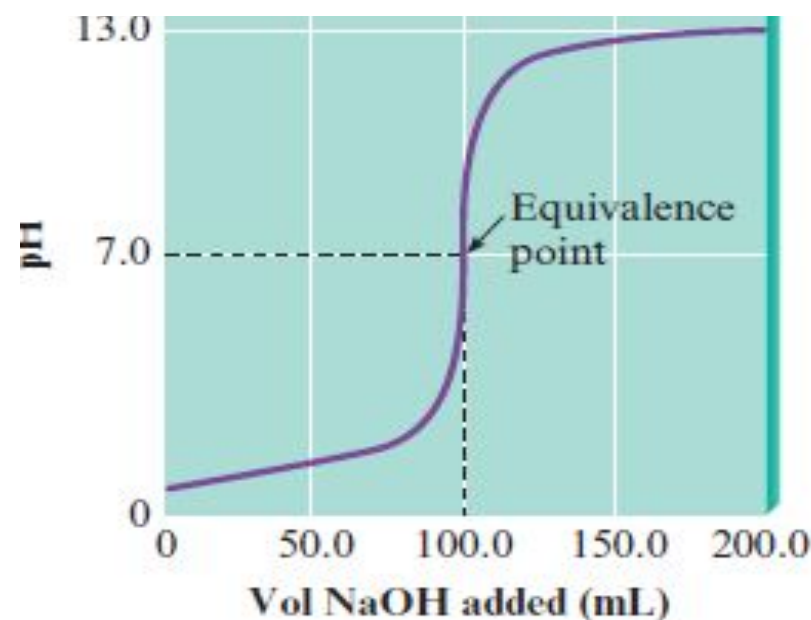


The pH curve in Fig. 15.1, typical of the titration of a strong acid with a strong base, has the following characteristics:

Before the equivalence point, $[H^+]$ (and hence the pH) can be calculated by dividing the number of millimoles of H^+ remaining by the total volume of the solution in milliliters.

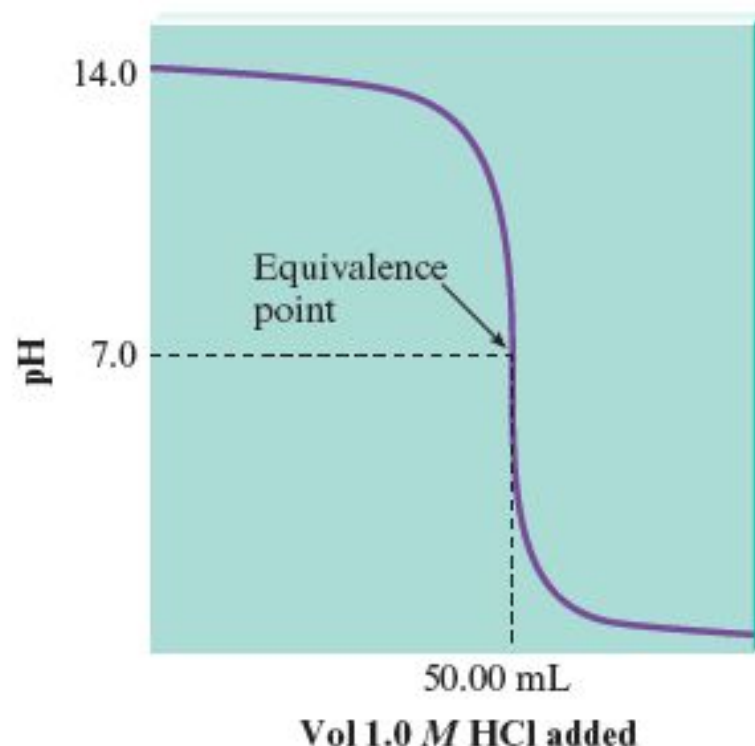
At the equivalence point, the pH is 7.00.

After the equivalence point, $[OH^-]$ can be calculated by dividing the number of millimoles of excess OH^- by the total volume of the solution. Then $[H^+]$ is obtained from K_w .



The titration of a strong base with a strong acid requires reasoning very similar to that used above, except, of course, that OH^- is in excess before the equivalence point and H^+ is in excess after the equivalence point. The pH curve for the titration of 100.0 mL of 0.50 *M* NaOH with 1.0 *M* HCl is shown in Fig. 15.2.

Figure 15.2 | The pH curve for the titration of 100.0 mL of 0.50 *M* NaOH with 1.0 *M* HCl. The equivalence point occurs at 50.00 mL HCl added, since at this point 5.0 mmol H^+ has been added to react with the original 5.0 mmol OH^- .



Acid-base indicator

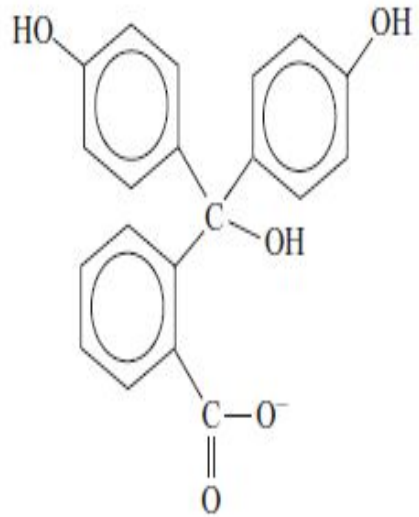
An acid-base indicator is an organic substance used for the detection of equivalence point or neutral point in an acid-base titration. An indicator has one colour in acid solution and entirely different in basic solution.



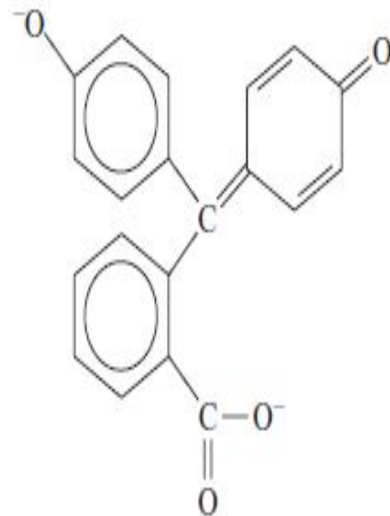
The indicator phenolphthalein is colorless in acidic solution and pink in basic solution.

The most common acid–base indicators are complex molecules that are themselves weak acids (represented by HIn). They exhibit one color when the proton is attached to the molecule and a different color when the proton is absent. For example, phenolphthalein, a commonly used indicator, is colorless in its HIn form and pink in its In[−], or basic, form. The actual structures of the two forms of phenolphthalein are shown in

15.5 Acid–Base Indicators

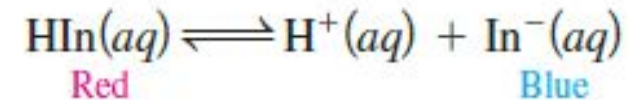


Colorless acid form, HIn



Pink base form, In[−]

To see how molecules such as phenolphthalein function as indicators, consider the following equilibrium for some hypothetical indicator HIn, a weak acid with $K_a = 1.0 \times 10^{-8}$.



$$K_a = \frac{[\text{H}^+][\text{In}^-]}{[\text{HIn}]}$$

By rearranging, we get

$$\frac{K_a}{[H^+]} = \frac{[In^-]}{[HIn]}$$

Suppose we add a few drops of this indicator to an acidic solution whose pH is 1.0 ($[H^+] = 1.0 \times 10^{-1}$). Then

$$\frac{K_a}{[H^+]} = \frac{1.0 \times 10^{-8}}{1.0 \times 10^{-1}} = 10^{-7} = \frac{1}{10,000,000} = \frac{[In^-]}{[HIn]}$$

This ratio shows that the predominant form of the indicator is HIn, resulting in a red solution. As OH^- is added to this solution in a titration, $[H^+]$ decreases and the equilibrium shifts to the right, changing HIn to In^- . At some point in a titration, enough of the In^- form will be present in the solution so that a purple tint will be noticeable. That is, a color change from red to reddish purple will occur.